PART I LIFE AND DEATH

TWO

From Prehistory to 1945

THE WORLD IS A HEALTHIER PLACE now than at almost any time in the past. People live longer, they are taller and stronger, and their children are less likely to be sick or to die. Better health makes life better in and of itself, and it allows us to do more with our lives, to work more effectively, to earn more, to spend more time learning, and to enjoy more and better time with our families and friends. Health is not a single quantity like temperature; someone might have terrific eyesight but poor physical stamina, or might live for many years but have serious recurrent depression or migraines. The seriousness of any given limitation depends on what a person does or would like to do. My lousy throwing arm was an occasional embarrassment in high school, but it is not a problem for a professor. Health has many dimensions, and it is hard to boil it down to a single convenient number. However, one aspect of health is easy to measure and is of overriding importance: the simple fact of being alive or dead. Knowing this is of limited use for an individual—one would certainly expect more from one's physician than the diagnosis "well, you are alive"—but measures of life and death are invaluable for thinking about the health of groups of people, either whole populations or subgroups, such as men and women, blacks and whites, or children and the elderly.

One familiar measure of life and death is how long a newborn baby can expect to live. This is known as life expectancy at birth, or often simply life expectancy. Provided life is worth living, having more years to live is good, and it is usually true (but not inevitably so) that populations in which people live longer are also populations in which people are healthier while they are alive. We saw in Chapter 1 how life expectancy varies around the world, how it is longer in richer countries, and how it has been generally increasing over time. In this chapter, we look in more detail at the how and why of life expectancy and how the world got to where it is now. This book is not a history of health, nor a history of life expectancy,[1] but there is much to be learned from looking at the past, and we will not have much chance of a better future if we do not try to understand it.

To establish where we are now, and to introduce some of the ideas that we shall need, I start with the past century or so of mortality and life expectancy in the United States. I then retreat back—way back—to see what life was like at its very beginnings, and then fast-forward to about 1945. The end of World War II is a convenient resting point because, after 1945, there are much better data, and because the story line changes.

Basic Notions of Life and Death, Illustrated by the United States

Life expectancy in the United States increased from 47.3 years in 1900 to 77.9 years in 2006. Figure 1 plots the numbers separately for men and women; women typically live longer than men, and they did so throughout the twentieth century. Both men and women had large increases in life span: 28.8 years for men and 31.9 years for women. The rate of increase was faster in the first half of the century but continues on today; in the past quarter-century, the life span of men has increased by one year every five years, and the life span of women, by one year every ten years. The first thing to take away from the figure, as is the case for much of this book, is that things are getting better, and hugely so. To have increased life spans by thirty years over a little more than a century is an extraordinary achievement, a Great Escape indeed. Having registered that big fact, we can note some of the secondary features of the figure. Why are men and women so different, not only in the years of life they can expect, but also in the rates at which their life expectancies are improving? Why does the first half-century look so different from the period after World War II?

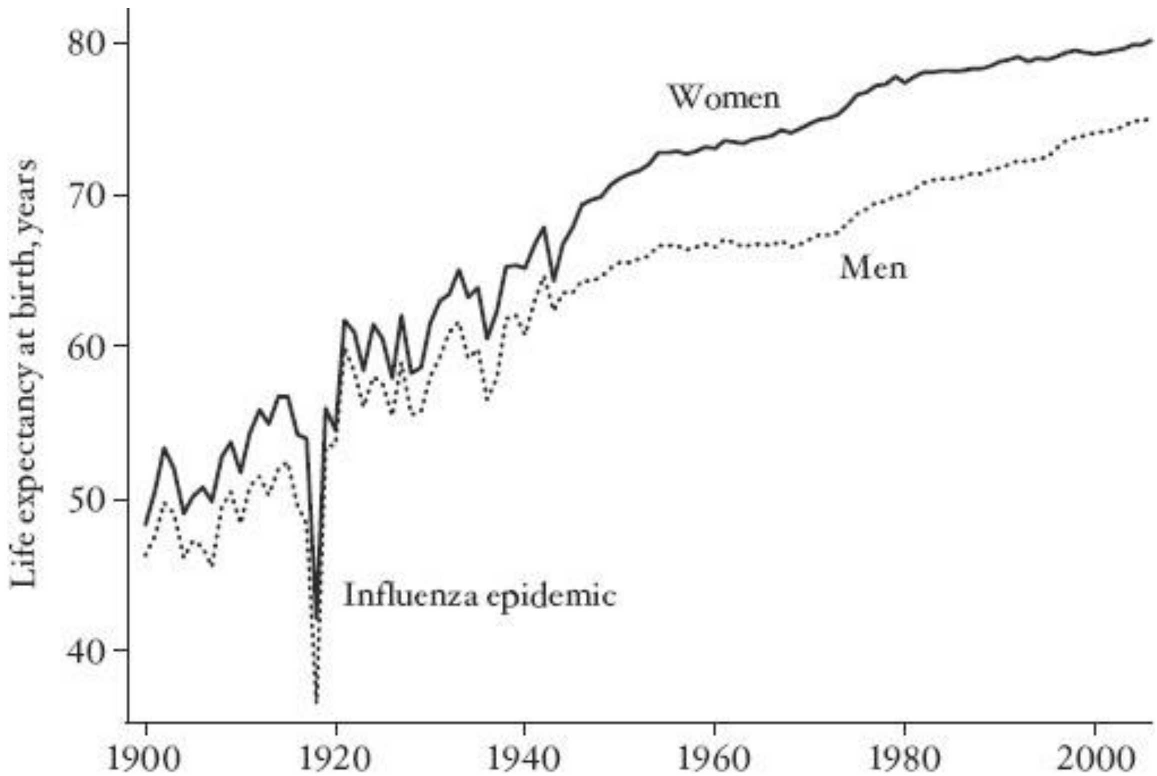

FIGURE 1 Life expectancy for men and women in the United States.

An immediate feature of the figure is the large dip in life expectancy during the influenza epidemic at the end of World War I. Life expectancy in 1918 was 11.8 years lower than life expectancy in 1917 and then rose again by 15.6 years in 1919, and life expectancy was back on its trend immediately after the epidemic. In the world as a whole, more than fifty million people may have died in the epidemic, more than half a million of them in the United States. Yet the way that life expectancy is defined tends to exaggerate the epidemic's effect on the life chances of a newborn child. In hindsight, we know that the flu epidemic would last only one year, so that if the baby made it through its first year of life, there would be no further threat from the epidemic. But when demographers calculate life expectancy in 1918, they assume that the epidemic will be permanent, and in 1919, they forget that it ever existed. This may seem like an odd way to assess life chances, but, in fact, it is hard to see how to do better.

When we are presented with a newborn child today and asked to calculate how long we think he or she will live, we need to know about the risks of dying in the years ahead, something that we do not know. Demographers work around that problem by using information on the risks at the time of birth, and they calculate how long the baby would be expected to live if the risks of dying at each age were exactly the same as they are today. In the case of an epidemic like the 1918–19 influenza epidemic, the mortality risk at every age was suddenly increased in 1918, so when life expectancy at birth is calculated for that year, the assumption is that the baby will face the age-specific risks of the epidemic in every year of life. This would make sense if the epidemic were to last forever, or at least for the lifetime of the child, but if the epidemic lasts only a year or two, the sharp drop in life expectancy overstates the real risks over the child's life. We can do better than this, but only with hindsight, once we wait until everyone in the child's birth cohort has died—which will be a wait of more than a century—or if we make forecasts. But forecasts have difficulties of their own—for example, no one in 1917 would have predicted the influenza epidemic at all.

Standard life expectancy measures—the ones that do not wait until everyone is dead and that do not make forecasts—are called "period" measures, meaning that they are calculated on the assumption that the mortality risks of the period are fixed forever. Not only can this be an issue for episodes like the influenza epidemic but it is also a problem for thinking about life expectancy today. When we look at Figure 1 and think about the future, it is hard not to think that life expectancy will continue to increase and mortality rates will continue to fall. Which means that life expectancy today, which is a little over 80 for a girl child born in the United States, is likely to underestimate the expectation of life for today's newborn, who might reasonably expect—if progress continues—to live to be a centenarian.

The influenza epidemic is only one of the reasons why the graphs in Figure 1 are so much more variable before 1950 than after 1950. While there was nothing comparable to that disaster, there were many smaller waves of disease that were large enough to show up in the life expectancy of the population. Infectious diseases that do not concern us much today were still threats in the United States in 1900, when the leading causes of death were, in order of importance, influenza, tuberculosis, and diarrhea. Tuberculosis remained in the top three until 1923, and in the top ten until 1953. Infectious diseases, including pneumonia, diarrheal disease, and measles, brought early death to many children. At the beginning of the century the deaths of children from these infectious diseases were relatively more important than they are today, when most deaths are among the elderly, and from chronic diseases like cancer and heart disease, not from infections. This change is the same epidemiological transition that we saw in Chapter 1 when we compared rich and poor countries today, and which took place over time in today's rich countries.

The "aging of death," from children to the elderly, makes life expectancy less sensitive to year-to-year fluctuations in deaths which, with the reduction in infectious disease, are themselves less pronounced today than in the past. Saving the lives of children has a bigger effect on life expectancy than saving the lives of the elderly. A newborn who might have died but does not has the chance to live many more years, which is not the case when a 70-year-old is pulled through a life-threatening crisis. This is also one of the reasons why the rate of increase in life expectancy has slowed down in recent years; mortality among children is now so low that progress can only really take place among older adults, among whom reductions in mortality rates have smaller effects on life expectancy.

Just because life expectancy is more sensitive to early-life than to late-life mortality, it does not follow that it is more important or more worthwhile to save the life of a child than to save the life of an adult. That is an ethical judgment that depends on many factors. On the one hand, saving a child saves many more potential years of life, while on the other, the death of a newborn does not entail the end of the many projects, interests, relationships, and friendships that are part of an adult life. Along these lines, the economist Victor Fuchs has suggested that the value of a life might be judged by the number of people who come to the funeral, a not entirely serious proposal that neatly captures the idea of down-weighting the very young as well as the very old. But such matters cannot be settled by the mechanical choice of a particular measure of health, such as life expectancy. Life expectancy is a useful measure, and it captures much that is important about population health. However, if we choose it as a measure of wellbeing, and set it as one of society's targets, we are buying into an ethical judgment that more heavily weights mortality at younger ages. Such judgments need to be explicitly defended, not simply adopted without thought.

The choice of life expectancy can sometimes be downright misleading. Figure 1 shows that life expectancy rose much more rapidly in the first half of the twentieth century than in the second half. This happened because infant and child mortality was high in 1900, and because reductions in mortality among the young have much larger effects on life expectancy than the reductions in mortality among the middle-aged and elderly that were so important at the end of the century. If we think of life expectancy as *the* measure of population health, or even as a good measure of general social progress, we can easily convince ourselves that the United States did better before 1950 than after 1950. It is certainly possible to argue that case, but focusing on life expectancy prioritizes mortality decline among the young over mortality decline among the old, and that is an ethical choice that needs to be argued, not just taken for granted. The same issue appears when we compare mortality decline in poor countries—mostly among children—with mortality decline in rich countries—mostly among the elderly. If we use life expectancy, poor countries are catching up in health and welfare, but such a catch-up is not a *fact* about health or even about mortality in general but an *assumption*, that life expectancy is the best indicator of health and social progress. I shall return to these issues in Chapter 4.

Figure 1 shows that the difference in life expectancy between American men and women, although always in favor of women, is different today than it was in the past. The difference in life spans was two to three years at the beginning of the twentieth century and rose in fits and starts until the late 1970s, after which it fell again, in the early years of the twenty-first century, to about five years. The differences in mortality rates between males and females are far from fully understood. Women face lower risks of death than men throughout the world, and throughout life; men are at higher risk even before they are born. The exception is maternal mortality, a risk from which men are exempt, and the reduction in maternal mortality in twentieth-century America is one reason why women's life expectancy has risen faster than men's.

A much more important reason is changing patterns of smoking. Smoking causes death both through heart disease—relatively quickly—and through lung cancer—with about a thirty-year wait between exposure and death. The slowdown in the rate of improvement in life expectancy among men in the 1950s and 1960s owes much to the earlier rise in their cigarette smoking. Men started smoking much earlier than women—among whom smoking was socially disapproved for many years, an injustice that did wonders for women's health!—but men also quit much sooner. The slowdown in life expectancy growth for women happens right at the end of the graph, two or three decades after the corresponding slowdown for men. In recent years, American women have also sharply reduced their smoking, and the rates of lung cancer for women have begun to turn down, as did those for men many years ago. For the rich countries of the world in the second half of the twentieth century, smoking is one of the most important determinants of mortality and of life expectancy.

The inequality in mortality between men and women is far from the only inequality between groups in the United States. In 2006, life expectancy at birth for African-American men was six years less than life expectancy for white men; for women, the difference was in the same direction, but smaller: 4.1 years. And like the differences between men and women, these differences have not remained constant over time. The Centers for Disease Control and Prevention estimate that, at the beginning of the twentieth century, there was a more than fifteen-year gap in life expectancy between whites and what were then known as nonwhites, a broader category than African-Americans.

Inequalities in life expectancy echo other inequalities between black and white in America—in income, in wealth, in education, and for much of the century, even in the opportunity to vote or to run for office. This consistent pattern of inequality in so many dimensions means that the gaps in wellbeing are even starker than the gaps in any one dimension, such as mortality or income. Any study of inequities between blacks and whites in the United States has to look at the whole picture at once, not just at health or only at wealth. Inequalities in mortality between ethnic and racial groups are not well understood, although unequal provision of health care is certainly important. The reduction of gaps in life expectancy and in child mortality is part of the general reduction in racial differences over the century, and a reduction in one inequality tends to contribute to reductions in others. That such differences defy simple explanations is shown by the mortality rates of Hispanics in the United States, whose life expectancy in 2006 was two and a half years *longer* than the life expectancy of non-Hispanic whites. The Great Escape from premature death in the United States has come to both men and women, and to all racial and ethnic groups, but the different groups started from different places, and made their escapes at different rates, so that patterns of inequalities have also changed over time.

Although the United States spends almost twice as much on health care as any other country, its citizens are not the longest lived. British and American life expectancies were very similar until the 1950s. There then followed a twenty-year period of British advantage which was lost in the 1980s, but then opened up again in the late 1990s and the early years of the new century. A gap of less than half a year in 1991 had grown to a gap of one and a half years by 2006. The gap between the United States and Sweden is much larger, more than three years in favor of the Swedes. Although the Swedish advantage has grown in the past few years, it goes back as long as we have records. In Chapter 4, I shall return to the gaps in life expectancy between the rich countries and try to explain what might be causing them. As is the case between groups within the United States, the experience of escape has been different for different countries. As we shall see, these differences are dwarfed by the differences between rich and poor countries.

To understand more about life expectancy, we need to dig deeper and look at mortality at different ages. Figure 2 shows how mortality rates vary with age for a selection of countries and years: Sweden in 1751 (Swedish data go back further than in any other country), the United States in 1933 and in 2000, and the Netherlands, also in 2000.[2] (The graph for Sweden in 2000 is close to that for the Netherlands, but a little lower at young and high ages.) These graphs show mortality rates for each age up to age 80—the number of people older than 80 thins out to the point where the graphs are unreliable. Mortality rates are shown as deaths per thousand people who are alive at that age. So, for example, the top curve shows that in Sweden in 1751, more than 160 out of every thousand newborns did not make it to their first birthday, while at age 30, only ten out of every thousand 30-year-olds did not survive to become 31. The logarithmic scale is useful here too, and I have used it for the vertical axis, so that the (fourfold) move from 0.5 to 2 shows up as the same size as the (fourfold) move from 10 to 40. The lowest mortality rates in the graph, for 10-year-olds today, are a thousand times lower than the mortality rates for newborns in Sweden in 1751, or only a tenth of the rates for 10-year-olds in the United States in 1933.

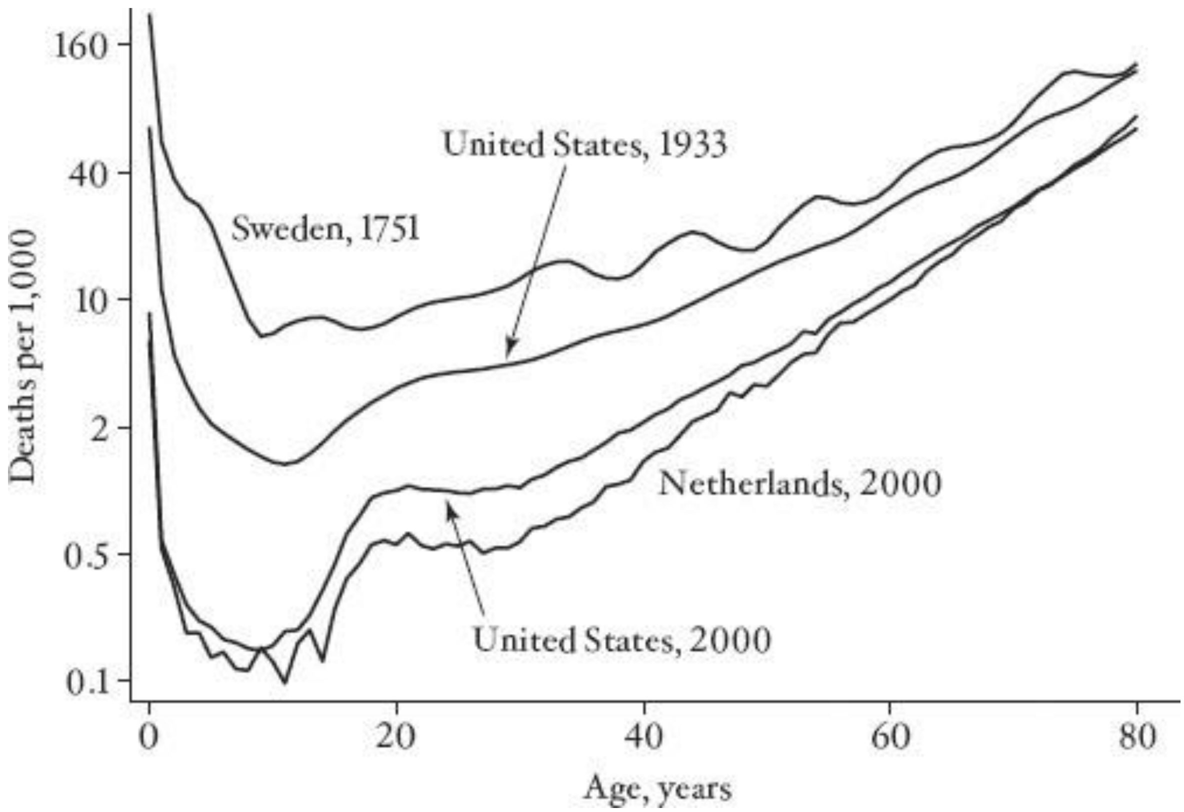

FIGURE 2 Mortality rates by age, selected countries and periods.

Mortality curves have a characteristic shape, reminiscent of the Nike “swoosh”: starting high at low ages, falling sharply to reach a low point in the early teens, and then rising steadily with age. The risk of death is at its greatest early in life and then again in old age. A vivid illustration comes from a sign in the washroom of a maternity hospital that I visited, urging patrons to wash their hands thoroughly because “the first few days of life are critical.” Beneath this was scrawled, “not as critical as the last few days.” The joke mostly pokes fun at the medical profession’s use of the word *critical,* but it neatly highlights the fact that we are in most danger of dying very early and very late in life.

Whether young life or old life is the more dangerous has changed over the years. In Sweden in 1751—well before the modern mortality decline—it was riskier to be a newborn than to be an 80-year-old. Today, when the chance of dying in the first year of life is less than 1 percent, it is more than six times riskier to be an 80-year-old. In the eighteenth century and for thousands of years before that, many people died as children; in Sweden in 1751, about a third of children died before their fifth birthday. Today, in Sweden and in other rich countries, almost everyone lives to die when they are old; indeed the Swedish infant mortality rate today is only around three per thousand.

The changing balance between young and old mortality means that, in a country where many children die, almost no one will actually live the number of years that national life expectancy says they should. We usually think of an average as a sort of "representative" or typical figure, but one of the peculiarities of the average number of years lived is that this is incorrect. In Sweden in the late eighteenth century, life expectancy was in the low to mid-30s, from which one might easily, but wrongly, conclude that few people lived to be old, and that few children ever got to know their grandparents. But this was not true; if you were lucky enough to get past the dangers of childhood, then you had a good chance of making it into old age—not as good as would be the case today, but enough to ensure that you had a fair chance of knowing your grandchildren. An extreme case would be one in which half of newborns die at birth and the other half live to be 50. Life expectancy at birth is 25, but no one dies at 25, and remaining life expectancy at first birthday is 49 years, 24 years longer than life expectancy at birth! A less extreme but real example comes from England in the middle of the nineteenth century, where life expectancy at 15 (sometimes called "adult" life expectancy) was higher than life expectancy at birth; I will come back to this later. More generally, keeping the mortality "swoosh" in mind is the key to understanding changes in life chances over time as well as the differences between rich and poor countries.

The mortality "swooshes" in Figure 2 show steady progress over time, with the "swooshes" for later dates always beneath those for earlier dates. We do not have data for the United States or the Netherlands in the eighteenth century, but we can suppose that the picture was roughly similar to that for Sweden. Life in 1933 and in 2000 was much less risky, with very large percentage reductions in mortality compared with earlier years, especially at younger ages, but not excluding the elderly, who did particularly well between 1933 and 2000. The comparison of the Netherlands and the United States in 2000 shows, once again, that the United States does badly compared with other rich countries; mortality rates in the United States in 2000 were higher than mortality rates in the Netherlands at all ages up to age 73. This pattern between the United States and the Netherlands carries over to comparisons between the United States and other rich countries. For those who live long enough, mortality rates are extraordinarily low in the United States, perhaps because of the willingness of the American medical system to use every means available to save lives, even among those with only a few years left to live.

The bottom two curves, for the United States and the Netherlands in 2000, show a temporary peak in mortality around age 20. Between ages 15 and 34, the leading causes of death are not disease—except briefly during the AIDS epidemic and before the advent of antiretroviral drugs—but accidents, homicide, and suicide. The mortality curves for earlier periods show that these dangerous and sometimes deadly behaviors by young people—particularly young men—are much more pronounced now than seventy years ago, and they do not appear at all in Sweden in the eighteenth century.

Where do the numbers in the figures come from? How do we know about mortality rates? In the wealthy countries of the Organisation for Economic Co-operation and Development (OECD) today, all births and deaths are registered with the government as they occur. Babies have birth certificates, and when people die, doctors or hospitals issue death certificates listing their particulars, including age, sex, and cause of death. This is known as the "vital registration system," *vital* in this context meaning to do with life and death. In order to make sure that birth and death records are correct, the vital registration system needs to be *complete,* meaning that every birth and every death must be registered. To get mortality rates, we also need to know the population by age, sex, and race so as to be able to calculate the fractions who have died; these counts come from regular censuses of the population, which most countries carry out every decade or so (for some reason, almost always in years ending in either zero or one).

Sweden was one of the first countries to have a complete vital registration system, which is why we have Swedish mortality rates for as early as the eighteenth century. London started collecting “bills of mortality” in the seventeenth century, and parish registers in Europe go back even further. The Puritans in Massachusetts thought that registration should be the business of the state, not the church, and Massachusetts had a vital registration system by 1639. Yet it was only in 1933 that all American states had complete registration, itself an important marker of government capacity. Without comprehensive data on births and deaths, a society is ignorant of the most basic facts about its citizens, and many of the roles that government exercises, and that today we take for granted, become impossible. The Swedes in the eighteenth century and the Puritans in Massachusetts were visionaries and pioneers in good government.

The American life expectancy data in Figure 1 before 1933 refer only to the states with registration. For countries with less than complete vital registration, or that lack good census data—probably a majority of countries in the world even today lack the state capacity to maintain either—demographers have developed tricks and approximations for filling in the blanks. For infant and child mortality, which remains a common occurrence in many countries, surveys of mothers can tell us how many children have been born and how many have survived. The United States Agency for International Development funds an invaluable series of surveys—the Demographic and Health Surveys—which collect this information for many poor countries where vital registration either does not exist or exists but is ignored in practice. (Parents do not register the births of their children, and when children or adults die, they are buried or cremated according to local custom, without the information being gathered into any national database.)

For the deaths of adults, there remain substantial gaps in information in many countries—where even the best estimates are little more than guesses—and in those cases it is impossible to draw the complete mortality "swooshes" that appear in Figure 2. Life expectancy is a little easier to guess, because it is so heavily influenced by child mortality, but in countries where adult mortality is unusual or variable—such as those affected by the HIV/AIDS epidemic—life expectancy estimates also need to be treated very cautiously. For all of these reasons, it is useful to look at the health experience of the poorest countries separately from that of the rich countries, which is what I shall do in Chapters 3 and 4.

Life and Death in Prehistory

How did today's mortality patterns come about? What caused the huge increase in life expectancy in the twentieth century? What was life like in the past, what made it get better, and what lessons does the past hold for improving the health of the large fraction of the world's population that has not yet escaped from the tentacles of early death?

For perhaps 95 percent of the time that humans have existed, for hundreds of thousands of years, people lived by hunting and gathering. Today, when there are only a few hunter-gatherer groups left in the world, nearly all living in marginal environments such as deserts or the Arctic, it might seem strange that life among such people has any relevance for our health. But it was the hunter-gatherer existence that shaped us, if only because of the enormous spans of time involved. Human beings *evolved* to be hunter-gatherers, and our bodies and minds were adapted for success in such an existence. Humans have lived in modern circumstances, as farmers or city-dwellers, for "only" a few thousand years, and it helps in understanding our health today if we know something about the conditions for which our bodies were designed.

We cannot look back and know how our ancestors lived and died hundreds of thousands of years ago, but much has been learned from the archeological record, including the examination of skeletal remains (paleopathology), which provide an amazing amount of information about nutrition, disease, and causes of death. Paleopathology can also estimate age at death from even partial skeletons, so we know something about life expectancy. Anthropologists have been studying actual hunter-gatherer groups for much of the past two hundred years, though some of the best evidence—including medical evidence—comes from contemporaneous groups (with appropriate adjustments made for their contact with modern society). Taken together, the two sources of evidence have provided a remarkable amount of useful data.[3]

Diet is a good starting point. So is exercise. Hunter-gatherers did a lot of brisk walking, tracking down prey, perhaps ten or fifteen miles a day. Their diet contained mostly fruits and vegetables, which were typically more easily obtained than animals. Wild plants—as opposed to their cultivated descendants—are fibrous, so hunter-gatherers ate lots of roughage. Meat was highly prized but often scarce, although some of the most fortunate groups lived in times or places where large wild animals were plentiful. Meat from wild animals has a much lower fat content than the meat from the farmed animals we eat today. People ate a wide variety of plants and meat, more so than in many agricultural communities even today, so micronutrient deficiencies were rare, as were diseases associated with them, such as anemia. Work was a cooperative activity, carried out with family and friends, and people depended on others to be successful in obtaining food. All of this sounds just like what my doctor tells me at each annual physical: get more exercise; eat less animal fat, more fruits and vegetables, and more roughage; and spend less time by yourself in front of a screen and more with your friends having fun.

Although hunter-gatherers knew nothing of modern hygiene, their behaviors helped protect their health, at least to some extent. Fertility was low by the standards of the poorest countries today, with the average woman giving birth to about four children, widely spaced and breast-fed for long periods. Low fertility may have been aided by infanticide, but breast-feeding—which lowers the probability of conception—would also have helped, as perhaps did the fact that women, like men, got lots of exercise. The contamination of food or water by human excrement—what in polite circles is called the fecal-oral transmission route for disease—is an efficient way of passing infection from one person to another and would kill millions in later eras. The fecal-oral link is obviously less dangerous where population density is low, and many groups of hunter-gatherers did not stay in one place long enough for accumulated waste to become an unmanageable threat. Even so, around 20 percent of children died before their first birthday, a number high by modern standards but not very different from, and in many cases better than, that seen in now-rich (but then-poor) countries in the eighteenth and nineteenth centuries—not to mention a number of poor countries in the twentieth and twenty-first.

Exactly how hunter-gatherers were organized depended on where they lived and on the local environment. But we can imagine a hunter-gatherer band containing thirty to fifty individuals, many of whom were kin, and small enough for everyone to know everyone else well. The band might be linked with other such bands within broader networks of hundreds, or in some cases thousands, of individuals. Within the band, resources were shared remarkably equally, and there were no leaders, kings, chiefs, or priests who got more than their fair share or who told other people what to do. According to one account, anyone who tried to set themselves above others was laughed at and, if the behavior persisted, killed.[4] One reason why equal sharing might have been important is that most of the groups did not or could not store food. So if one hunter and his friends had successfully hunted a woolly mammoth (or a lizard weighing a ton, or a 400-pound flightless bird) they could eat until they could eat no more, but they would have no way of keeping the leftovers for days when no mammoth, lizard, or bird was to be found. One good workaround is to share the mammoth with the whole group, so that when someone else kills another large animal on another day, last month's mammoth-catchers will get a share too. Over hundreds of thousands of years, individuals and groups that were good at sharing would do better than individuals and groups that did not share, so that evolution could eventually produce a species with a hardwired belief in sharing. Our current deep-seated concerns with fairness, as well as our outrage when our norms of fairness are violated, are quite possibly rooted in the absence of storage options for prehistoric hunters. There is even some evidence that in places where limited storage was possible—in northern as opposed to equatorial latitudes—societies tended to be more unequal.

Hunter-gatherer societies were egalitarian societies that managed without rulers, but we should not regard them as paradises, as Gardens of Eden before the Fall. Encounters with other groups were frequently violent, sometimes to the extent of ongoing warfare, and many men died in battle. Because there were no leaders, there was no effective system of law and order, so that internal violence—often between men fighting over women, or as a result of disagreements—went unpoliced, another source of high adult mortality. Hunter-gatherers were exempt from some infectious diseases, though others, like malaria, have likely been present throughout human history. Small groups cannot maintain infectious diseases, such as smallpox, tuberculosis, or measles, that confer (sometimes limited) immunity upon recovery, but they are subject to zoonotic diseases whose normal hosts are wild animals or the soil, as well as to a range of parasites such as worms. Life expectancy at birth among hunter-gatherers, around 20–30 years depending on local conditions, was short by today's standards, although not by historical standards in the West, nor within living memory in countries that remain poor today.

The availability of food varied from place to place and time to time, so that there would have been inequality between groups, and the wealth and longevity of groups would have changed over time. There is skeletal evidence suggesting periods of plenty, particularly in places where there was an abundance of easily hunted large animals—buffalo in the American West or large flightless birds in Australia. In these places and times, hunter-gatherer groups were what the anthropologist Marshall Sahlins described as the original affluent societies.[5] Large wild animals provided a rich and well-balanced diet—their fat content was only 10 percent of that of artificially fed and rarely exercised modern farm animals—and could be killed with minimal work, so that people in such groups had a high material standard of living and lots of leisure. Yet this Garden of

Eden, if such it was, was lost as many of the large animals were hunted to extinction, forcing people to switch to plants and seeds and to smaller, harder-to-catch animals like rodents. This prehistoric degradation reduced the standard of living, and the skeletons of people from this era—who got less to eat from childhood on—are shorter than those of their more fortunate predecessors.

The history of the wellbeing of hunter-gatherers—their nutrition, their leisure time, and their mortality rates—is important for the general themes of this book. We should not suppose that the wellbeing of mankind has steadily improved over time or that human progress has been universal. We spent most of our history as hunter-gatherers, and during that time, as food got scarcer and work got harder and longer, life got worse, not better. Worse was to come as people moved from foraging to farming. Although we are now used to better (where "we" refers to the privileged inhabitants of today's rich world), the capability to live such long and good lives is a recent gift that even now has not been bestowed on everyone in the world. The anthropologist Mark Nathan Cohen, whose *Health and the Rise of Civilization* is one of my main sources here, concludes his review by observing that "the undeniable successes of the nineteenth and twentieth centuries have been of briefer duration, and are perhaps more fragile, than we usually assume."[6]

We also learn from this distant past that inequality has *not* characterized all human societies. For most of history there was no inequality, at least within groups of people who lived together and knew each other. Inequality, instead, is one of the "gifts" of civilization. Again, quoting Cohen, "The very process that creates the potential of civilization simultaneously guarantees that the potential is unlikely to be aimed equally at the welfare of all of its citizens."[7] Progress in prehistory—like progress in recent times—is rarely equally distributed; a better world—if indeed a world with agriculture *was* a better world—is a more unequal world.

The invention of agriculture—the Neolithic revolution—began "only" about ten thousand years ago, a brief period indeed compared with the hunter-gatherer era that preceded it. We are accustomed to thinking of "revolutions" as transformative *positive* events—the Industrial Revolution and the germ-theory revolution are the two obvious examples. Yet it is unclear that agriculture was an advance to a higher plateau of wealth and health, as opposed to a retreat from an older way of living that had become unsustainable as animal stocks and suitable plants ran out under pressure of rising numbers and rising temperatures at the beginning of the Holocene. Like the "broad-spectrum" revolution that preceded it—the switch from large animals to small animals, plants, and seeds—the turn to agriculture is perhaps more accurately seen as an adaptation to the increasing difficulty of foraging for food, as was argued many years ago by the economist Esther Boserup.[8] Agriculture may have made the best of a bad job, and giving up foraging for a sedentary life as a farmer may have been better than living on increasingly hard-to-find and ever-smaller wild seeds, but it should not be seen as part of a long-term trend to better wellbeing. Hunter-gatherers who had access to wild game, worked little, and thoroughly enjoyed the hunts were unlikely to have voluntarily swapped their lives for the drudgery of agriculture and what *The Communist Manifesto* called the "idiocy of rural life." Morris summarizes Sahlins's argument as, "Why did farming ever replace foraging if the rewards were work, inequality, and war?"[9]

Settled agriculture allowed food to be stored, in granaries and in the form of domesticated animals. Agriculture allowed and was made more efficient by the possession of property and the development of priests and rulers, towns and cities, and within-community inequality. Larger settlements and the domestication of animals brought new infectious diseases, such as tuberculosis, smallpox, measles, and tetanus. The Neolithic revolution probably did little to increase life expectancy and may actually have reduced it, if only because children continued to die in large numbers, from malnutrition and from germs, as well as from the new diseases, and because sanitation is harder to deal with and fecal-oral transmission harder to prevent in large, sedentary communities. Immobile agricultural communities also limited the diversity of food, and domesticated crops are in many cases less nutritious than their wild progenitors; stored food can also be spoiled food and yet another source of disease. Trade between communities could offset the monotony of the local menu, but it also brought new threats of disease. These "new" diseases, transmitted from previously unconnected civilizations, brought infections against which local populations had no immunity; they could and did cause huge mortality, up to and including the collapse of whole communities and civilizations.[10]

There is no evidence of any sustained increase in life expectancy for thousands of years after the establishment of agriculture. It is possible that adult mortality rates fell somewhat as child mortality rates rose; with very high mortality among children, those who survive may be particularly hardy. Women in agricultural settlements had more children than their foraging ancestors, and although they also lost more, the switch to agriculture helped population numbers increase. In good times, or when productivity increased through innovations, the new possibilities led not to sustained increases in per capita income or life expectancy, but rather to increased fertility and expansions of population as the carrying capacity of the land increased. In bad times, in famines or epidemics, or when there were more people than could be fed, population decreased. This Malthusian equilibrium persisted for millennia. Indeed, it is possible that the decline in individual wellbeing that took place toward the end of the foraging period continued long after agricultural settlement, albeit with interruptions, right through until the last 250 years.

We are so used to thinking of progress in terms of rising incomes and extended lives that we can easily make the mistake of ignoring the increase in wellbeing that comes from simply having more people. If it is true that having more people in the world implies that there will be less for each person—because of diminishing returns, for example—then the highest possible per capita wellbeing would come about in a world with only one person—hardly what we think of as a good world. Philosophers have debated these issues for many years; one position, argued by the philosopher and economist John Broome, is that once people are above some basic subsistence point that makes life worth living, then having more such people makes the world a better place.[11]

The world is supporting more total wellbeing. If so, and provided that life was worth living for most people—admittedly a large proviso—the long Malthusian era from the invention of agriculture up to the eighteenth century should be regarded as a period of progress, even if living standards and mortality rates showed no improvement.

Life and Death in the Enlightenment

Fast-forward a few thousand years to a period for which we begin to have good data on mortality. The British historical demographer Anthony Wrigley and his colleagues have reconstructed the history of English life expectancy from the parish registers that recorded the births, marriages, and deaths (hatches, matches, and dispatches) of the population.[12] These parish records are not as good as a vital registration system—the study covered only a sample of parishes, there are issues with people moving from one parish to another, newborns who died very soon after birth may not have shown up at all, and parents sometimes reused the names of such children—but they provide by far the best record that we have for any country before about 1750. The line in Figure 3 shows the estimates of the life expectancy of the general population of England from the middle of the sixteenth century to the middle of the nineteenth century. Although there are sharp fluctuations from year to year associated with epidemics—smallpox, bubonic plague, and the "sweating sickness" (possibly influenza, possibly some other virus that no longer exists)—there is no clear trend over the three hundred years of the reconstruction.

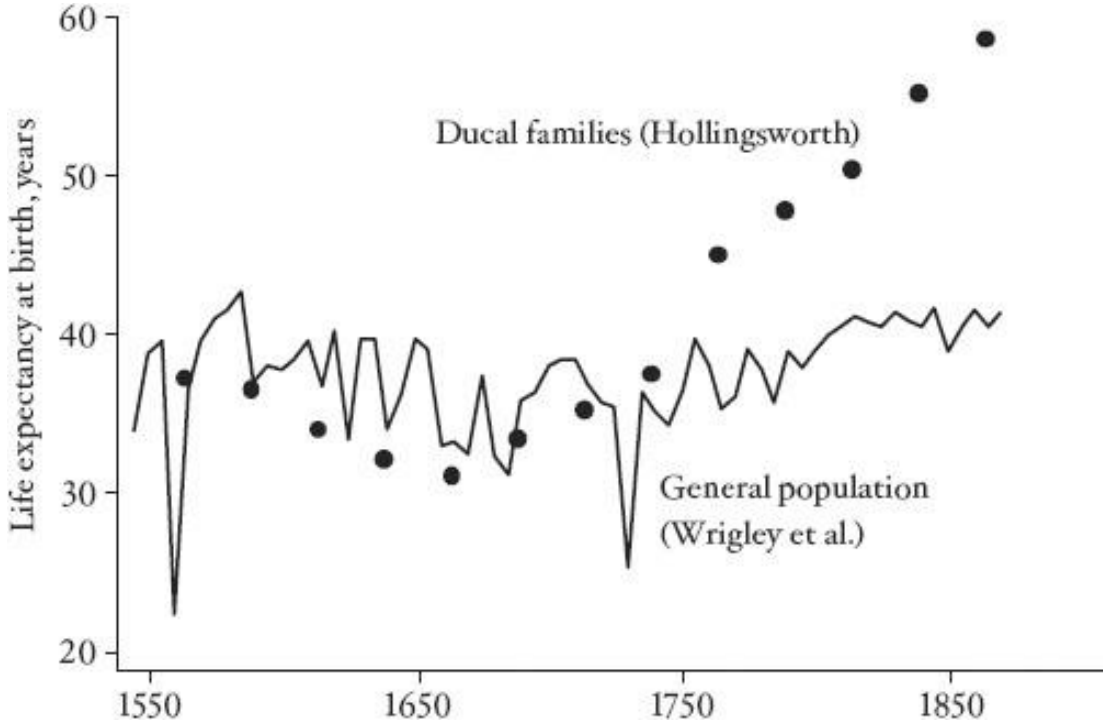

FIGURE 3 Life expectancy for the English population and for ducal families. (After Bernard Harris, 2004, "Public health, nutrition, and the decline of mortality: The McKeown thesis revisited," *Social History of Medicine* 17(3): 379–407.)

The circles in the figure show the life expectancy of the English aristocracy for each decade of the same three centuries; these data were assembled by the historical demographer T. H. Hollingsworth in the 1960s from the typically meticulous records of births and deaths kept by the British peerage.[13] The idea of superimposing the peers on the populace comes from the social historian Bernard Harris, who first drew this wonderfully informative diagram.[14] From 1550 to about 1750, the life expectancy of the dukes and their families was similar to, or perhaps a little lower than, that of the general population. This in itself is perhaps surprising; richer and higher-status populations often have better health than poorer and lower-status populations, a phenomenon that is known as the health "gradient"; there is evidence of this as far back as Ancient

Rome. So the first lesson is that this “gradient” in health is not universal and was not present in Britain for at least two centuries.

There is little doubt that the British aristocrats got more to eat than did the common people; courtiers of Henry VIII at Hampton Court consumed 4,500 to 5,000 calories a day in the sixteenth century, and the king himself eventually became so obese that he could not move without assistance. Henry was not alone, and in some other European courts people consumed even more.[15] Yet more food—or at least more food of the kind that the aristocrats consumed—did nothing to protect against the bacteria and viruses that brought plague and smallpox, or from the poor sanitation that did away with their children. So the comparison with the peerage suggests that, in England from 1550 to 1750, it was disease, not lack of nutrition, that set the limits to life expectancy. Of course, disease and undernutrition compound one another—it is hard to digest food when you are sick—but there is no evidence that the consistently high nutrition levels of the aristocracy protected them or their children against the infectious diseases of the day.

After 1750, the life expectancy of the aristocracy pulled away from that of the general population, opening up a nearly twenty-year gap by 1850. After about 1770, there is also some upward movement in life expectancy for everyone. Looking at this figure alone, the movement looks comparable to other ups and downs since 1550, but it is significant in hindsight because of what was to happen after 1850—a sustained increase in life expectancy for the whole population, an increase that continues to this day. Life expectancy at birth in England and Wales was to rise from 40 years in 1850 to 45 in 1900, and almost 70 by 1950. The aristocracy not only opened up a health gradient in the last half of the eighteenth century, they also got a head start on the general increase in life expectancy that was to come.

We do not know for sure *why* the gap opened up, but there are many good guesses. This was the British Enlightenment, summarized by the historian Roy Porter as a time when people stopped asking “How can I be saved?”—a question that over the past century had brought little but mayhem, including a civil war—and asked instead “How can I be happy?”[16] People began to seek *personal* fulfillment, rather than seeking virtue through obedience to the church and “performing the duties appropriate to one’s place in society.”[17] Happiness could be pursued by using reason to challenge accepted ways of doing things, including obedience to the crown and the church, and by finding ways of improving one’s life, including both material possessions and health. Immanuel Kant defined the Enlightenment by the mottoes “Dare to know! Have the courage to use your own understanding.” During the Enlightenment, people risked defying accepted dogma and were more willing to experiment with new techniques and ways of doing things. One of the ways in which people began to use their own understanding was in medicine and fighting disease, trying out new treatments. Many of these innovations—in this earlier age of globalization—came from abroad. The new medicines and treatments were often difficult to obtain and expensive, so that, at first, few could afford them.

Inoculation for smallpox, or variolation, was one of the most important of these innovations.[18] Smallpox was a leading cause of death in Europe in the eighteenth century. In cities that were large enough for the disease to be permanently present, almost everyone caught smallpox in childhood, and those who survived had lifetime immunity. The inhabitants of towns and villages would often escape the disease for many years, but would have no immunity during an epidemic, and large numbers of both children and adults died. In Sweden in 1750, 15 percent of all deaths were due to smallpox. In London in 1740, there were 140 smallpox burials—mostly of children—for every 1,000 baptisms in the city.

Variolation is not the same as vaccination, which was developed by Edward Jenner only in 1799, was widely and rapidly adopted thereafter, and is credited with major reductions in mortality. Variolation was an ancient technique, practiced in China and India for more than a thousand years, and also long established in Africa. Material was extracted from the pustules of someone suffering from smallpox and scratched into the arm of the person to be protected; in the

African and Asian versions, dried scabs were blown into the nose. The inoculee developed a mild case of smallpox but became immune thereafter; according to the History of Medicine Division of the U.S. National Institutes of Health, only 1–2 percent of those variolated died, compared with 30 percent of those exposed to smallpox itself.[19] The technique has always been controversial, and it is likely that some of those who were variolated could spread smallpox to others, and perhaps even start a full-fledged epidemic. No one would advocate the practice today.

The introduction of variolation in Britain is credited to Lady Mary Wortley Montague, who, as wife to the Turkish ambassador, had seen the practice in Constantinople and pressed for its adoption in Britain at the highest levels of society. Impressed, members of the royal family were variolated in 1721, although not before some condemned prisoners and abandoned children were pressed to serve as guinea pigs, variolated, and subsequently exposed to smallpox without ill effect. Variolation then spread widely among the aristocracy. The historian Peter Razzell has documented how, over the next three-quarters of a century, variolation started out as a very expensive technique—involving several weeks of isolation and substantial charges by the inoculators—and eventually became a mass campaign that inoculated ordinary people. Local authorities even paid to have paupers inoculated, because it was cheaper to inoculate them than to bury them. By 1800, the number of smallpox burials per baptism in London had fallen by half.

In the United States, variolation crossed the middle passage in the slave ships; the population of Boston was completely inoculated by 1760, and George Washington inoculated the soldiers of the Continental Army. Smallpox epidemics in Boston had killed more than 10 percent of the population in the late 1600s and in 1721, when variolation was first tried, but there were relatively few smallpox deaths after 1750.

The late eighteenth century saw other health and medical innovations, described by the medical historian Sheila Ryan Johansson.[20] Cinchona bark (quinine) was first introduced to Britain from Peru as a treatment for malaria, “holy wood” (guaiacum) was brought from the Caribbean and used as a treatment for syphilis (supposedly more effective, and certainly more expensive, than mercury), and ipecac was brought from Brazil as a treatment for “the bloody flux.” Professional (male) midwives were used for the first time by the wealthy, an innovation imported from France. This was also the time of the first public health campaigns (for example, against gin), the introduction of the first dispensaries, and the beginnings of city improvement. In my home town of Edinburgh in Scotland, a New Town was built starting in 1765; the old city was not destroyed, but its central and heavily polluted North Loch was drained, and a new, spacious, and salubrious town was built to the north. Sir Walter Scott, who was born in the old town in 1771, lost six of his eleven brothers and sisters in infancy, and he himself contracted polio as a child, yet his family could not be described as impoverished; his mother was the daughter of a professor of medicine and his father, a solicitor.

We have no way of quantifying the effects of these innovations on mortality, and even the one most likely to have had the largest impact—variolation—remains controversial. Yet there is a plausible case that these innovations—all the result of better scientific knowledge, and born of the new openness to trial and error—were responsible for the better health of the peerage and the royal family at the end of the seventeenth century. At first, because they were expensive and not widely appreciated, they were confined to those who were wealthy and well informed, so that new inequalities in health were opened up. But those inequalities also signaled that general improvements lay just ahead, as the knowledge spread more widely, as the medicines and methods became cheaper, and as they led to new and related innovations that could cover the whole population, such as vaccination against smallpox after 1799 or the sanitarian movement that cleaned up the cities. We shall meet other examples of new knowledge opening up health inequalities that presaged general benefits, including the spread of the germ theory of disease at the end of the nineteenth century and the understanding of the health effects of cigarettes after the 1960s.

From 1800 to 1945: Nutrition, Growth, and Sanitation

If the improvements in life expectancy in the eighteenth century were muted and unequally distributed, no one could have missed the enormous and general improvements at the end of the nineteenth century and beginning of the twentieth. Figure 4 shows the progress of life expectancy for England and Wales, Italy, and Portugal; the data start earlier in Britain, with Italy following around 1875 and Portugal only in 1940. There are earlier data for the Scandinavian countries, and for France, Belgium, and Holland, but they would not be easily distinguishable from England on this graph. As we shall see, it is no accident that the countries that led the fight against mortality are those with the best and earliest data.

FIGURE 4 Life expectancy from 1850: England and Wales, Italy, and Portugal.

I focus on the English story here, but the diagram also highlights the diffusion of innovations, something we shall meet again and again. The English experience after 1850 is followed by other countries that got a later start (here Italy and Portugal), and what are initially very wide gaps in life expectancy—ten years between Italy and England in 1875 and about the same between England and Portugal in 1940—diminish over time, so that by the end of the twentieth century Italy had actually overtaken England, and Portugal was not far behind. As was the case for the peers and the people in England at the end of the eighteenth century, whatever happened in England—and shortly thereafter in the countries of northern and northwestern Europe, the United States, and Canada—generated a gap between them and the countries of southern (and eastern) Europe, as well as the rest of the world. Over time, those gaps have narrowed, as progress has spread and become more generalized—not evenly, not everywhere, and not completely, but eventually to the whole world. A better world makes for a world of differences; escapes make for inequality.

So what did happen in England? What caused life expectancy to double, from 40 years to almost 80 years, over a century and a half? Given the long history of thousands of years of stable or even falling life expectancy, this is surely one of the most dramatic, rapid, and favorable changes in human history. Not only will almost all newborns live to be adults, but each young adult has more time to develop his or her skills, passions, and life, a huge increase in capabilities and in the potential for wellbeing. Yet, perhaps surprisingly, this greatest of benefits is still less than fully understood and was little researched until late in the twentieth century.

A good starting point is life expectancy, not at birth, but at age 15, sometimes called adult life expectancy; this is defined as the number of *additional* years a 15-year-old can expect to live, calculated in just the same way as life expectancy at birth, but starting, not from zero, but from 15. Figure 5 shows life expectancy at birth as in Figure 4 (though I have taken the opportunity to switch to the total population, including armed forces, so that the mortality in World War I makes the dip in 1918 even larger), as well as life expectancy at age 15. At 15, people could expect to live for another 45 years in 1850, compared with 57 years a century later in 1950.

What is most notable about Figure 5 is that, until around 1900, adult life expectancy in Britain was actually *higher* than life expectancy at birth. In spite of having lived for 15 years, these teenagers could expect a longer future than when they were born. Because life as an infant or child was so dangerous, life expectancy shot up once you survived childhood. By the end of the twentieth century, the chances of dying in childhood have become very small—at least in the rich countries—so that the gap between adult life expectancy and life expectancy at birth has expanded, and it is now almost the full fifteen years that it would be if no one died before their fifteenth birthday. These patterns are similar in other countries for which we have data, although the date at which life expectancy at birth overtook adult life expectancy varies from country to country: up to ten years earlier in Scandinavia and ten to twenty years later in Belgium, France, and Italy.

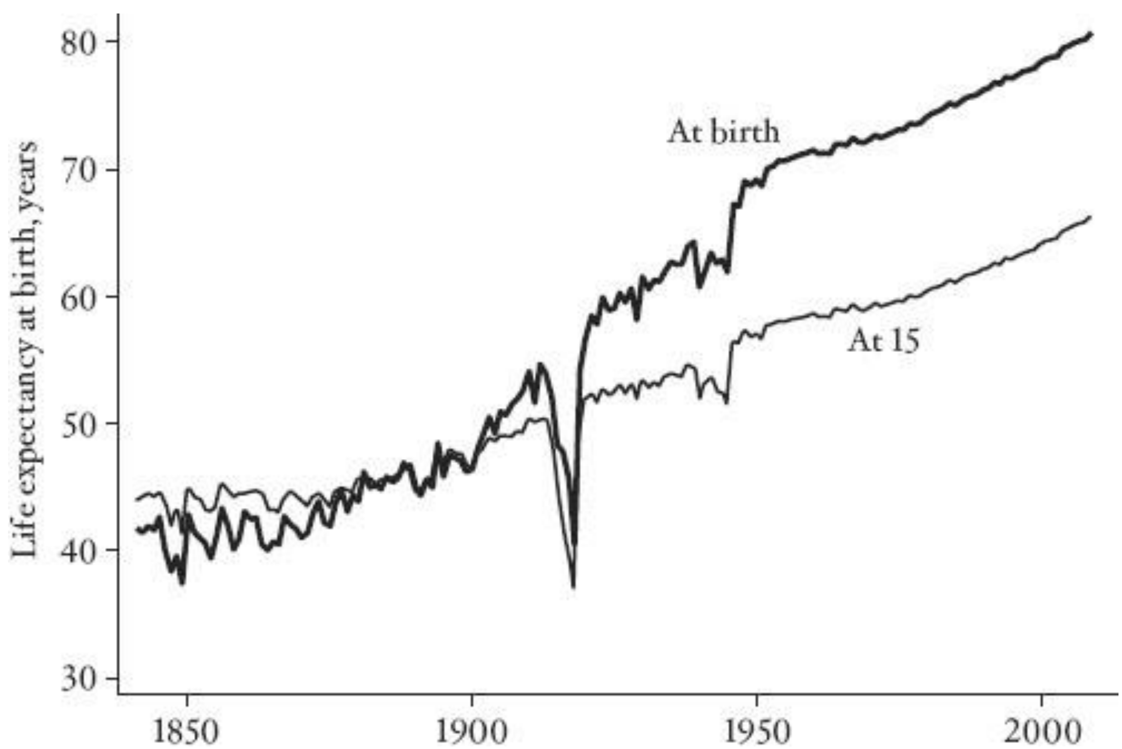

FIGURE 5 Life expectancy at birth and at 15: England and Wales, total population.

Whatever caused the increase in life expectancy from 1850 to 1950 operated most powerfully by lowering the chances that children died. Factors that reduced mortality for adults, or factors that worked for both adults and children, were also important, but less dramatic in their effect.

Decreased child mortality cannot have had much to do with medical treatments, such as new medicines or drugs such as antibiotics, sulfa drugs, or streptomycin for tuberculosis, in part because most of the decrease in mortality took place long before such treatments were available, and in part because the introduction of the drugs did not result in any sharp changes in mortality from the diseases that they treat. The founder of social medicine, the Englishman Thomas McKeown, drew a series of famous diagrams showing for a whole series of diseases that mortality rates were falling *before* the introduction of the effective treatment, and continued to fall at much the same rate *after* its introduction.[21] McKeown, a physician himself, concluded that medicine was not very useful (he even argued that the higher the status of a doctor, the more useless he or she was likely to be)[22] and concluded that the roots of health improvement lay in economic and social progress, particularly in better nutrition and living conditions. McKeown was the first in a long line of physicians who came to feel that their own professional efforts could do little to improve public health, and who turned to more general social ills, such as poverty and deprivation, which they saw as the fundamental causes of poor health. McKeown thought that gradual improvement in the material conditions of life, such as better food and better housing, were much more important than either health care or even public health measures. McKeown’s views, updated to modern circumstances, are still important today in debates between those who think that health is primarily determined by medical discoveries and medical treatment and those who look to the background social conditions of life.

Nutrition was clearly part of the story of early mortality decline. The population of Britain in the eighteenth and early nineteenth centuries consumed fewer calories than they needed for children to grow to their full potential, and for adults to maintain healthy bodily functioning and to do productive and remunerative manual labor. People were very skinny and very short, perhaps as short as at any previous (or subsequent) time. Throughout history, people adapted to a lack of calories by not growing too big or too tall. Not only is stunting a consequence of not having enough to eat, especially in childhood, but smaller bodies require fewer calories for basic maintenance, and they make it possible to work with less food than would be needed by a bigger person. A six-foot-tall worker weighing 200 pounds would have survived about as well in the eighteenth century as a man on the moon without a spacesuit; on average there simply was not enough food to support a population of people of today's physical dimensions. The small workers of the eighteenth century were effectively locked into a nutritional trap; they could not earn much because they were so physically weak, and they could not eat because, without work, they did not have the money to buy food.

With the beginnings of the agricultural revolution, the trap began to fall apart. Per capita incomes began to grow and, perhaps for the first time in history, there was the possibility of steadily improving nutrition. Better nutrition enabled people to grow bigger and stronger, which further enabled productivity to increase, setting up a positive synergy between improvements in incomes and improvements in health, each feeding off the other. When the bodies of children are deprived of the nutrients they need to grow, brain development is also unlikely to reach its full potential, so these larger, better-off people may also have been smarter, further adding to economic growth and speeding up the virtuous circle. Taller, bigger people lived longer, and better-nourished children were less likely to die and better able to ward off disease. This account has been developed over the years by the Nobel laureate economist Robert Fogel and his collaborators.[23]

There is no doubt that nutrition has improved and that people have become bigger, stronger, and healthier. But focusing entirely on food cannot provide a complete account of the decline in child mortality. Such an approach underplays the importance of the direct control of disease, and it focuses too much on the unaided role of the market economy and too little on the collective and political efforts that were behind the control of disease. The economist and historian Richard Easterlin has argued convincingly that, when we try to match the onset of economic growth with the improvements in health, the timing is wrong.[24] The improvements in child mortality across northwestern Europe were much too uniform to be explained by economic growth, which began at different times in different countries; we shall later see echoes of this in the twentieth century in the international synchronization of improvements in heart disease. And if food itself was so important, why did the British aristocrats, who had plenty of food, not do better than the ordinary people in the centuries before 1750? The demographer Massimo Livi-Bacci has documented similar cases in several European countries, including monasteries of well-fed monks, who ate rich and varied diets but died at the same rate as everyone else.[25] Food may protect against some diseases, but it is far from a universal prophylaxis. It may perhaps protect better against bacterial than viral diseases, but it is not clear that even that idea is correct.

The major credit for the decrease in child mortality and the resultant increase in life expectancy must go to the control of disease through public health measures. At first, this took the form of improvements in sanitation and in water supplies. Eventually science caught up with practice and the germ theory of disease was understood and gradually implemented, through more focused, scientifically based measures. These included routine vaccination against a range of diseases and the adoption of good practices of personal and public health based on the germ theory. The improvement of public health required action by *public* authorities, which required political agitation and consent and could not have been accomplished through the market alone, although rising real incomes certainly made it easier to fund often costly sanitary projects. At the individual level, the reduction in diseases—particularly diarrheal, respiratory, and other infections among children—improved nutrition and helped to account for the increases in height, strength, and productivity. Food intake is important, but more important is *net* nutrition, the amount of nutrition that is actually available to people after allowing for nutrition lost to disease, directly in the case of diarrhea, but also to fighting fevers and infections. The improvements in sanitation, followed by measures based on the germ theory of disease, were the major factors in improving life expectancy in northwestern Europe and in the British offshoot countries in the century after 1850. They spread to southern and eastern Europe in the early twentieth century, and eventually, after World War II, to the rest of the world, a development I shall discuss in the next chapter.[26]

The Industrial Revolution in Britain brought millions of people from the countryside to new cities like Manchester, where there were new livelihoods in manufacturing but little or no provision for dealing with the health hazards posed by so many people living in close quarters. Rural living can be reasonably safe in the absence of formal arrangements for the disposal of human wastes, but the same is not true in the cities. Domestic animals, horses for transport, cows for milk, and pigs for garbage disposal and food often lived in close proximity to their owners in the new cities. There were also dangerous wastes from factories and “nuisance” processes like tanning and butchery, and drinking water was often contaminated by human and other wastes. There were more public latrines in Ancient Rome than in Manchester during the Industrial Revolution.[27] When the same sources that provided drinking water were used to dispose of feces, the fecal-oral link that had been a problem since the Neolithic revolution was amplified to industrial strength. Life expectancy in the cities—as is still the case in some poor countries today—fell far below life expectancy in the countryside. Indeed, migration to the unhealthy cities helps explain why life expectancy of the general population was so slow to rise at the beginning of the nineteenth century, and why the general increase in life expectancy was postponed until after 1850. Eventually, these stinking and dangerous cities, the “dark, satanic, mills,” provoked a public reaction beyond pronouncements about the sad moral state of the sufferers, and local authorities and public health officers began to provide public sanitation.

The sanitarian movement had no new science to guide its efforts. Indeed its theory of disease, the "filth theory" or the "miasma theory"—that if it smelled bad it was bad for health—was wrong and was no different from what public health authorities had believed in Italy when (mostly unsuccessfully) fighting the Black Death in the fourteenth century. Yet there was sufficient truth to the theory to make it effective if rigorously pursued. People are indeed less likely to get sick if human wastes are safely disposed of and if the city water does not smell bad. But the theory led to too much emphasis on sanitation and not enough on water supplies, so that, at one point, the health authorities in London were emptying the stinking cesspools in basements into the Thames, thus recycling cholera into the water supply. A few years later, in London's cholera epidemic of 1854, one of the two water companies supplying the city with drinking water from the Thames had its inlets downstream of the sewage discharges, recycling cholera bacteria from one generation of victims to the next. Indeed, the fact that the other major company had recently moved its inlet to purer water upriver enabled John Snow, then a physician in London, to map the cholera deaths and match them to the offending water company, and thus to demonstrate that cholera was spread through contaminated drinking water.[28] This was one of the first "natural experiments" in public health, and it gets my vote as one of the most important of all time. Yet Snow recognized that the experiment was hardly decisive—for example, it might have been that one water company might have served only well-to-do patrons, who were protected for other reasons—and went to great pains to rule out other potential explanations for his results.[29]

Snow's findings, together with the later work of Robert Koch in Germany and Louis Pasteur in France, helped establish the germ theory of disease, albeit with much resistance from holdout believers in miasma theory. One sticking point was why some people exposed to the disease did not become sick—a serious challenge to causality and understanding.[30] Indeed, Koch, who had isolated *Vibrio cholerae* in 1883, proposed four "postulates," all of which had to be satisfied if a microbe were to be safely identified as the cause of a disease. One was that, if the microorganism were introduced into a healthy person, the disease should follow. This gap in the theory was spectacularly demonstrated in 1892 when a prominent disbeliever and miasmatist, Max von Pettenkofer, then aged 74, publicly drank a flask of cholera bacteria, specially sent by Koch from Egypt, and suffered only mild negative after-effects. Exactly why he should have escaped is unclear—it was not stomach acidity, which he had neutralized—but many disease agents work only under suitable conditions, and von Pettenkofer had a theory of this kind, that the microorganism must first be converted to a miasma by putrefying in soil. This theory was proved tragically wrong in the Hamburg cholera epidemic in 1892; the neighboring city of Altona, which like Hamburg drew its water from the Elbe, filtered its water while Hamburg did not, and it escaped the epidemic. The swallowing of the bacillus came after the Hamburg epidemic and was something of a last act of defiance; von Pettenkofer shot himself in 1901.[31]

The discovery, diffusion, and adoption of the germ theory were the keys to the improvement in child mortality in Britain and around the world. The story also illustrates a number of themes that we shall meet again. Here was something new that had great potential for improving the wellbeing of mankind, in this case by saving children who would otherwise have died. The basic knowledge—that germs cause disease and, in the case of cholera, that the bacteria were spread through contaminated water—was free and available to anyone in the world without charge. Yet that did not mean that the policy measures that followed from the theory were adopted immediately or even quickly. For one thing, and as we have seen, not everyone was convinced. And even when people *were* convinced, there were all sorts of barriers. The knowledge might have been free, but adopting it was not. Constructing safe water supplies is cheaper than constructing sewage plants, but it is still costly, and it requires engineering knowledge as well as monitoring to ensure that the water is indeed uncontaminated. Sewage needs to be disposed of in a way that it does not pollute the supply of drinking water. Monitoring of individuals and of businesses is often difficult and often resisted, and it requires state capacity and competent bureaucrats. Even in Britain and the United States, the fecal contamination of drinking water was an issue well into the twentieth century. Turning the germ theory into safe water and sanitation takes time and requires both money and state capacity; these were not always available a century ago, and in many parts of the world they are not available today.

As always, there is the important matter of politics. The historian Simon Szreter describes how, in the cities of the Industrial Revolution, fresh water was widely available, but to factories as a source of power, not to the inhabitants of the cities to drink.[32] As is so often the case, the benefits of the new ways of doing things were far from equally distributed. And the factory owners, who were also those who paid taxes, had no interest in spending their own money on clean water for their workers. Szreter documents how new political coalitions of working men and displaced landholders successfully agitated to install the infrastructure for clean water, an agitation that was effective only after the Reform Acts enfranchised working men. Once the political balance had changed, the factory owners climbed on board, and cities began to compete with one another in advertising their healthfulness. (Princeton University, where I teach, did likewise at the same time, claiming that its elevation—all of 140 feet above sea level—made it a healthier environment for young men than the malarial swamps nearby.) Whenever health depends on collective action—whether through public works, the provision of health care, or education—politics must play a role. In this case, the (partial) removal of one inequality—that working people were not allowed to vote—helped remove another inequality—that working people had no access to clean drinking water.

Diffusion of ideas and their practical implementation take time because they often require people to change the way they live. Almost everyone in the rich world is now taught in school about the importance of germs and about how to avoid them by washing hands, by disinfection, and by the proper handling of foods and of wastes. But what we take for granted was unknown at the end of the nineteenth century, and it took many years before public and private behaviors could change to take full advantage of the new understandings.[33] The demographers Samuel Preston and

Michael Haines have described how, around the turn of the century, there were sharp differences in infant and child mortality rates between ethnic groups in New York City, so that, for example, Jews, whose religious observances were health-promoting, did much better than French Canadians, who had no such protection.[34] But the children of physicians died at much the same rate as the children of the general population until the germ theory of disease was understood, after which physicians' children were much less likely to die. In the United States, hotels did not change the bed linen from one guest to the next. At Ellis Island, physicians examined potential immigrants for trachoma (an infectious eye disease) using a buttonhook-like implement that was not sterilized between examinees; the immigration authorities were *spreading* trachoma, not halting it at the border.[35] A more contemporary example comes from India, where the *dai,* a traditional birth attendant, is often brought in to help women deal with complicated pregnancies. One such woman was observed by an American obstetrician, who marveled at her manipulative skill at reorienting an unborn child, a skill that would have made her rich in the United States. Yet this highly skilled professional never washed her hands between attending one woman and the next.[36]

Scientific advances such as the germ theory are not single discoveries but clusters of related discoveries, and they usually depend on earlier advances. Germs cannot be seen without microscopes, and although Anthony van Leeuwenhoek had made microscopes and used them to see microorganisms in the seventeenth century, such microscopes provided highly distorted images. In the 1820s, Joseph Jackson Lister developed the achromatic microscope, which used a combination of lenses to remove the distortions or "chromatic aberrations" that made earlier microscopes close to useless. The germ theory itself led to the identification of a range of causative microorganisms, including the bacteria for anthrax, tuberculosis, and cholera in Koch's laboratories in Germany. Koch was one of the founders of the then new field of microbiology, and his pupils went on to identify the microorganisms responsible for many diseases, including typhoid, diphtheria, tetanus, and bubonic plague. In the next wave of discovery, Louis Pasteur in

Paris demonstrated that microorganisms were responsible for the spoiling of milk and showed how to "pasteurize" milk to prevent it. Pasteur also showed how attenuated versions of infectious microorganisms could be used to develop a range of vaccines. (He also invented Marmite, a basic foodstuff without which life would be impossible for contemporary Britons; we shall meet it again in Chapter 6.) The germ theory also led Joseph Lister (son of Joseph Jackson Lister) to develop antiseptic methods in surgery which, together with the development of anesthetics, made modern surgery possible. The work of Snow, Koch, and Pasteur not only established the germ theory but also showed how to put it into practice for the public good.

Scientific advance—of which the germ theory is such a singular example—is one of the key forces leading to improvements in human wellbeing. Yet, as the gradual adoption of the germ theory demonstrates, new discoveries and new technologies are not enough without acceptance and social change. Nor should we think of scientific advances as appearing out of nowhere, like manna from heaven. The Industrial Revolution and the urbanization that accompanied it created the *need* for the scientific advance—people were dying of diseases that had not been a problem in rural England—but also the conditions under which it could be studied. The industrial-strength fecal-oral link that was forged by putting the effluent of one generation of cholera victims into the mouths and guts of the next provided an opportunity for *someone* to figure out what was going on. Of course, there is nothing inevitable about the process—the demand for cures does not always produce a supply of cures—but need, fear, and, in some circumstances, greed are great drivers of discovery and invention. Science develops according to the social and economic environments in which it exists, just as those environments themselves depend on science and knowledge. Even the microorganisms that play the central role in the germ theory do not exist in some pristine state waiting to be discovered. Their propagation, their evolution, and their virulence develop alongside the people that they infect. The conditions of the Industrial Revolution changed the conditions of life of millions of people, but they also changed the microorganisms that infected them, and the way in which they infected them, as well as creating the conditions under which the germ theory could evolve.

THREE

Escaping Death in the Tropics

FOR THE MAJORITY OF the world's population not fortunate enough to be born in a rich country, the battles against infectious disease had hardly been joined by 1945. Yet history did not have to be relived, or at least not at the same glacial pace. In 1850, the germ theory had yet to be established. By 1950, it was common knowledge, so that at least some of the improvements that had taken a century in the leading countries could happen more quickly in those that followed. That India today has higher life expectancy than Scotland in 1945—in spite of a per capita income that Britain had achieved as early as 1860—is a testament to the power of knowledge to short-circuit history. The rapid if uneven reduction in infant mortality in poor countries allowed millions of children to live who would otherwise have died and caused the "population explosion"—from 2.5 billion in 1950 to 7 billion in 2011—an explosion that is today gradually coming to an end. Over the postwar years, life expectancies in poor countries moved closer to life expectancies in rich countries, at least until the 1990s, when HIV/AIDS in Africa undid the postwar progress in the most seriously affected countries. Inequalities in life expectancy, which had expanded from 1850 when the rich countries pulled away, decreased after 1950 as poor countries caught up, and then expanded again with the advent of the new epidemic.

There are many countries where large fractions of children still die, and there are three dozen countries where more than 10 percent die before their fifth birthday. They are not dying of the "new" diseases, like HIV/AIDS, or exotic tropical diseases for which there is no cure. They are dying from the same diseases that killed European children in the seventeenth and eighteenth centuries, intestinal and respiratory infections and malaria, most of which we have known how to treat for a long time. These children are dying from the accident of where they were born, and they would not be dying had they been born in Britain, Canada, France, or Japan.

What is it that maintains these inequalities? What is it that makes it so dangerous to be born in Ethiopia or Mali or Nepal, and so safe to be born in Iceland or Japan or Singapore? Even in a country like India, where mortality rates have fallen rapidly, large fractions of children remain malnourished; they are skinnier and shorter than they ought to be for their age, and their parents are among the shortest adults on the planet, perhaps even shorter than the stunted adults in eighteenth-century England. Even today, and in spite of India being one of the fastest-growing countries in the world, why is it that so many Indians are trapped in the destitution that was the ultimate outcome of the Neolithic revolution?

In the years after World War II, in what the United Nations (UN) calls the less-developed regions of the world, large numbers of infants and children continued to die. In the early 1950s, more than a hundred countries lost more than a fifth of their children before their first birthday. These countries included all of sub-Saharan Africa, South Asia, and South East Asia. In 1960, the World Bank estimates that forty-one countries had child mortality rates (death by age 5) of more than a fifth, and in a few the rates were close to two-fifths. In the 1950s and 1960s, most of the world had mortality rates not very different from those in Britain a hundred or two hundred years before. But change was on the way.

The most rapid increases in life expectancy came very soon after the war. The demographer Davidson Gwatkin reports that around 1950, countries such as Jamaica, Malaysia, Mauritius, and Sri Lanka saw *annual* increases in life expectancy of more than one year for more than a decade.[1] In Mauritius, life expectancy rose from 33.0 years in 1942–46 to 51.1 years in 1951–53; in Sri Lanka, it rose by fourteen years in the seven years after 1946. Of course, these dashes for immortality cannot continue forever, and they can come only from large, one-time reductions in infant and child mortality. They were caused partly by the introduction of penicillin, which had first become available during the war; partly by use of the somewhat older sulfa drugs;

and probably in largest part by what is called "vector control," the chemical assault on disease-bearing pests, particularly mosquitoes and especially those of the genus *Anopheles*, which carry malaria. Much of the progress against malaria was later reversed when the mosquitoes became resistant and when the use of the highly effective insecticide DDT was stopped worldwide because of its environmental effects (largely from its overuse in agriculture in rich countries). Even if the effects on malaria were temporary, they were temporarily large, and subsequent advances in other directions, such as immunization campaigns, more than made up for the losses.

UNICEF, the arm of the UN responsible for the health and well-being of children, received the Nobel Peace Prize in 1965 for its work among the world's children. Immediately after World War II, UNICEF vaccinated children in Europe against tuberculosis, and it extended its reach in the 1950s to worldwide campaigns against tuberculosis, yaws, leprosy, malaria, and trachoma; it also sponsored clean water and sanitation projects. The Expanded Programme on Immunization (EPI) of the World Health Organization (WHO) was launched in 1974; it promoted immunization against diphtheria, pertussis (whooping cough), and tetanus (the DPT vaccine covers all three), as well as measles, polio, and tuberculosis. Most recently the Global Alliance for Vaccines and Immunisation (GAVI Alliance) was established in 2000, in an attempt to reinvigorate the work of the EPI. The progress of immunization has slowed somewhat in recent years, perhaps because the populations that were the easiest to reach and the most willing have already been covered. Another important innovation to help maintain the rate of mortality decline was the demonstration of the effectiveness of oral rehydration therapy (ORT) during a cholera outbreak in Bangladeshi and Indian refugee camps in 1973. A solution of salt and glucose in water, taken orally, prevents the dehydration that kills many children with diarrhea. The treatment costs only a few cents a dose, and it was hailed by the medical journal *The Lancet* as "potentially the most important medical advance this century."[2] ORT is another good example of how a pressing need, together with scientifically informed trial and error, can sometimes lead to a spectacular life-saving innovation.

These medical and technical advances were implemented even in places where local capacity was limited. Mosquitoes could be sprayed by foreign experts or contractors directed by foreign experts, and immunization campaigns could be directed from WHO in Geneva as short-term, almost military-style operations using local paramedics to give the shots. Vaccines were (and are) cheap and were often centrally obtained by UNICEF or WHO at favorable prices. These health campaigns, known as “vertical health programs,” have been effective in saving millions of lives. Other vertical initiatives include the successful campaign to eliminate smallpox throughout the world; the campaign against river blindness jointly mounted by the World Bank, the Carter Center, WHO, and Merck; and the ongoing—but as yet incomplete—attempt to eliminate polio.

Medical and public health advances were not the whole story; better education and higher incomes have helped too. Rates of economic growth have been high by historical standards since World War II, and there have been improvements in education—not everywhere, but in many countries. Women are more likely to be educated than used to be the case. In Rajasthan in India, where I was involved in collecting data, almost all of the adult women we interviewed could neither read nor write. Yet we regularly passed lines of uniformed girls (locally referred to by the British term “crocodiles”) setting off to school. Between 1986 and 1996, the fraction of rural Indian girls enrolled in school rose from 43 to 62 percent, and although the schools are sometimes terrible, even badly educated women are likely to be better and safer mothers than mothers who have no education at all. There is a large amount of research from India and other countries showing that the children of more educated mothers do better in both survival and subsequent outcomes; beyond that, educated women have fewer children and can devote more time and resources to each child. Lower fertility is good for mothers too, reducing the health risks of pregnancy and childbirth, and allowing women greater opportunities in their own lives.

Improvements in education may be the single most important cause of better health in lower-income countries today.

Economic growth puts more money into the hands of families, who are better able to feed their children, as well as into the hands of local and national governments, who are better able to make improvements in water supply, sanitation, and pest eradication. In most districts of India in 2001, more than 60 percent of households had access to piped water, while two decades before very few districts met this target; piped water is not always safe water, but it is much safer than water from most traditional sources.

Writing in 1975, the demographer Samuel Preston—the world's most acute observer of mortality—estimated that less than a quarter of the increase in life expectancy between the 1930s and the 1960s came from increases in domestic living standards, with the vast majority coming from new ways of doing things, vector control, new drugs, and immunizations.[3] Preston's calculations were for the limited group of countries for which he had data, several of which were not poor in 1945. His conclusion came from looking at graphs like Figure 3 in Chapter 1. He calculated how much life expectancy would have increased if the curve relating life expectancy to income had remained fixed and countries moved along it with economic growth (the contribution of income to better health), and how much of the gain came from the upward movement of the curve itself (the contribution of new methods that permit better health without any increase in living standards).

Later authors have split the credit between innovation and income differently, and there is no reason to suppose that the balance will be the same at all times, as Preston himself emphasized. The important new ways of saving lives—antibiotics, vector control, immunization—do not arrive evenly or predictably, and when one runs out of steam, there is no guarantee that there will be another waiting in the wings. Yet the big issues are always there: income on the one hand, treatment and innovation on the other hand, or the market versus public health, with education improving the effectiveness of both. If the diseases of poor countries are indeed "diseases of poverty" in the sense that they will vanish if poverty is reduced, then direct health interventions may be less important than economic growth. Economic growth would be "twice blessed"; it would increase material living standards directly *and* improve health as a bonus. If Preston's findings are still true today—a question I shall address later in this chapter—the magic of income will not be enough, and health must be addressed directly by health interventions. Note the similarity between Preston's findings and the conclusion of Chapter 2 that the mortality decline in Europe and North America from 1850 to 1950 was predominantly due to the conquest of disease by new ways of addressing health, with economic growth playing an important, but subsidiary, role.

Whatever takes the credit, there is no doubt about the extent of mortality reduction. The UN reports that, in the fifteen-year period from 1950–55 to 1965–70, the "less-developed regions" of the world saw an increase in life expectancy of more than ten years, from 42 to 53 years. By 2005–10, this had increased by another thirteen years, to 66 years. Although improvements continued in the "more-developed regions," they were much slower; see Figure 1, which shows the progress for selected regions of the world. The top line is for Northern Europe, defined as the Channel Islands, Denmark, Estonia, Finland, Iceland, Ireland, Latvia, Lithuania, Norway, Sweden, and the United Kingdom. In these countries together, life expectancy started at 69 and gained ten years by the beginning of the twenty-first century; I shall look at how this happened in the next chapter. The other regions, East Asia (including Japan), Latin America and the Caribbean, South East Asia, South Asia, and sub-Saharan Africa, have all gained more than 10 years, so that the gaps between them and Northern Europe have decreased. Even for sub-Saharan Africa, which has gained the least, the gap between it and Northern Europe has narrowed, from 31.9 years in the early 1950s to 26.5 years in 2005–10.

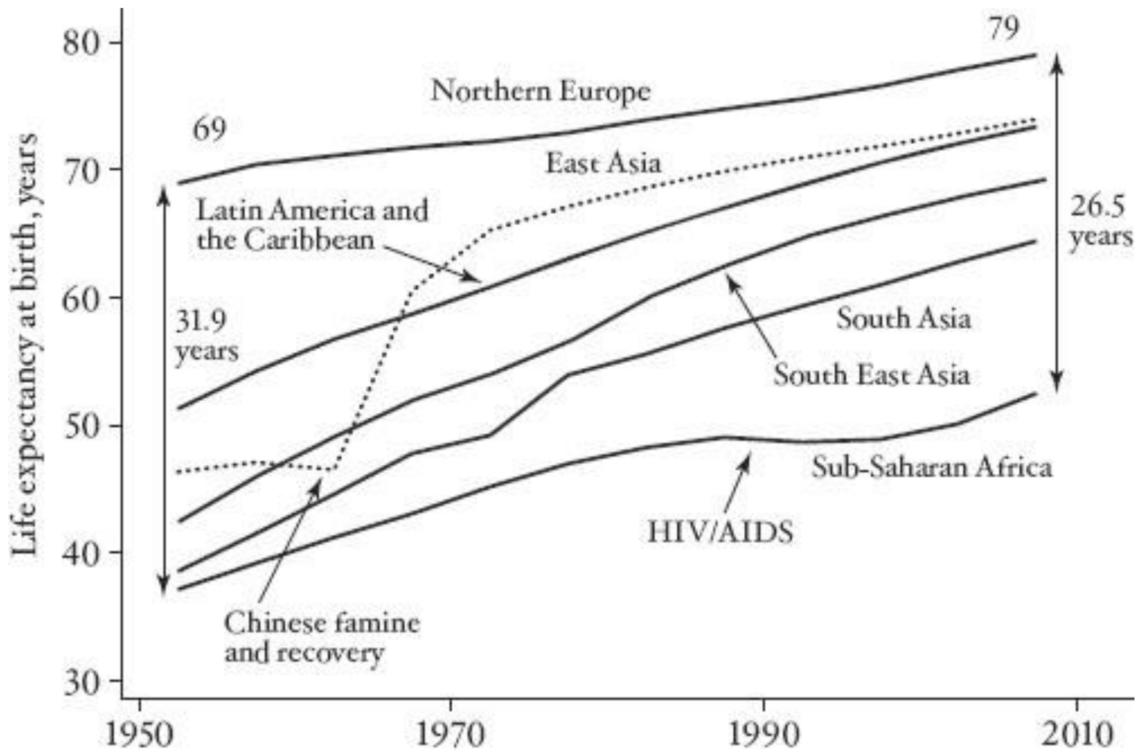

FIGURE 1 Life expectancy in regions of the world since 1950.

Africa and to a lesser extent South Asia (which extends as far north as Afghanistan) are the regions where the most remains to be done. Even before the HIV/AIDS epidemic, life expectancy in sub-Saharan Africa was growing more slowly than elsewhere, and HIV/AIDS caused a further stalling that is clearly visible in the figure. With the advent in recent years of antiretroviral therapy, and with behavioral change, the UN estimates that African life expectancy has begun to rise once again. Yet in the most affected countries most or all of the postwar progress was lost; life expectancy in Botswana—one of the best-governed and economically successful countries in Africa—rose from 48 years to 64 years and then fell back to 49 years in 2000–05, while Zimbabwe's life expectancy—in one of the worst-governed and economically unsuccessful countries in Africa—was lower in 2005–10 than in 1950–55. Great epidemics that kill millions of people—according to WHO, HIV/AIDS had killed thirty-four million by the end of 2011—clearly did not end after the influenza epidemic of 1918–19, nor should we be complacent about the absence of new epidemics in the future.

Although no one knows exactly how the AIDS epidemic began, the same cannot be said of the Chinese famine of 1958–61, whose origins I discussed in Chapter 1 and whose effects are clearly visible in Figure 1. As we shall see shortly, one-party rule in China can be capable of promoting public health by adopting measures that would sometimes face decisive opposition in democracies. However, when policies are disastrously wrong, there is likewise nothing to stop their implementation, even when the result is catastrophe. A contrast is often drawn between China, with its lack of democracy but effective policy implementation, and India, which is a democracy with a free press but often ineffective policies. Yet India has had no famine since its independence, although there were many under the British Raj.

In spite of the great setbacks from HIV/AIDS and the Chinese famine, Figure 1 shows that life chances are *better* than half a century ago in most of the world. But how good (or bad) is today's situation, and what remains to be done? A useful way to understand today's mortality is to look at deaths around the world—what people are dying from in countries at different levels of economic development—and try to understand which of these deaths might be avoided given what we know. If people are dying of the exotic and incurable "tropical" diseases that often appear in scare stories in the media, we need new cures and new medicines. If, in contrast, people are dying of the same old diseases that have long vanished from rich countries, we need to ask why people are still dying of things we know how to prevent. As we shall see, while there is certainly a need for new and better treatments, the major problem lies in the fact that too many of the world's children continue to die from what should be readily preventable diseases.

Table 1 gives the facts about global mortality in 2008 from WHO. These numbers involve a lot of estimation and should not be treated as accurate in detail, but the broad picture that they convey is reliable enough. The second column shows deaths for the world as a whole, the third for low-income countries, and the fourth for high-income countries. The division of the world by income comes from the World Bank, which divides the world into four categories: low income, lower middle income, upper middle income, and high income. Here I have shown only the top and bottom groups so as to focus on the inequalities in mortality between the richest and poorest. To give some idea of the countries involved, of the thirty-five low-income countries, twenty-seven are in Africa; the other eight are Afghanistan, Bangladesh, Cambodia, Haiti, Myanmar (Burma), Nepal, North Korea, and Tajikistan. India is no longer classed as a low-income country. There are seventy high-income countries, including most of the countries of Europe, North America, and Australasia; Japan; and a number of small oil-producing countries and a handful of island states.

TABLE 1

Global mortality in 2008, and in the poorest and richest countries

	World	Low-income	High-income
Percentages of deaths (percentages of population)			
Ages 0–4	14.6 (9)	35.0 (15)	0.9 (6)
Ages 60 and above	55.5 (11)	27.0 (6)	83.8 (21)
Cancer	13.3	5.1	26.5
Cardiovascular disease	30.5	15.8	36.5
Millions of deaths			
Respiratory infections	3.53	1.07	0.35
Perinatal deaths	1.78	0.73	0.02
Diarrheal disease	2.60	0.80	0.04
HIV/AIDS	2.46	0.76	0.02
Tuberculosis	1.34	0.40	0.01
Malaria	0.82	0.48	0.00
Childhood diseases	0.45	0.12	0.00
Nutritional deficiencies	0.42	0.17	0.02
Maternal mortality	0.36	0.16	0.00
From all causes	56.89	9.07	9.29
Total population	6,737	826	1,077

SOURCE: World Health Organization, Global Health Observatory Data Repository, downloaded February 3, 2013.

NOTES: Cardiovascular disease includes stroke. Respiratory infections are mostly lower respiratory infections (*lower* refers to infections below the vocal chords, including pneumonia, bronchitis, and influenza, which can also affect the upper respiratory tract). Perinatal deaths are deaths of children at birth or immediately thereafter and include deaths associated with babies being premature and of low birth weight, babies who die during birth, and babies who die from infections immediately after birth. Childhood diseases are whooping cough, diphtheria, polio, measles, and tetanus. About two-thirds of deaths from nutritional deficiencies are due to deficiency of protein or energy, and one-third are due to anemia.

The top part of the table shows how deaths divide up between children and the elderly, as well as the fractions that come from two of the leading noninfectious killers, cancer and cardiovascular disease. Deaths from cardiovascular disease include deaths attributable to diseases of the heart and of the veins, and so include strokes as well as heart attacks. The second column does the division for the world as a whole, the third and fourth for the low- and high-income countries. The bottom of the table shows raw counts in millions of deaths, focusing on the major killers in the low-income countries.

The top of the table shows in parentheses the percentages of the populations in each age group; the bottom of the table shows the population totals for each region. Note that most of the population of the world lives in the middle-income countries that are not shown here. The other key fact, in the top of the table, is that the low-income countries are much *younger* than the high-income countries. People have more children in poorer countries, and when populations are growing, each generation is larger than the previous one and the population is young. In some of the rich countries, baby boomers from the postwar years are now aging, which adds to the size of the 60-plus group. There are more than twice as many people aged 0–4 as people 60 and above in the low-income countries; in the high-income countries, there are more than three times as many elderly as children. Even if the risks were the same in poor and rich countries, there would be more deaths of children in the former, and more deaths of the elderly in the latter.

Infants and children account for 15 percent of all of the deaths in the world, while people aged 60 and over account for more than half. Yet that is not what happens in either poor countries or rich countries. In the poor countries, more than a third of deaths are of children under 5, and less than a third are of the elderly. In the rich countries, where the deaths of children are rare, more than 80 percent of deaths are of people 60 or older, and the vast majority of newborn children live to be old. In part, these differences are explained by the much larger fractions of old people in the rich countries, but not entirely—child deaths in relation to child populations are much higher in the low-income countries. The contrast between rich and poor comes from the epidemiological transition, according to which death itself "ages" as countries develop. The switch from death in childhood to death in old age also comes with a switch in the causes of death, from infectious disease to chronic disease. The fraction of people dying of cancer, stroke, and heart disease triples from low-income to high-income countries. In general, old people die of chronic disease, children of infectious disease.

The major killers in poor countries are largely the same diseases that used to kill children in the now-rich countries—lower respiratory infections, diarrhea, tuberculosis, and what WHO calls "childhood diseases": whooping cough, diphtheria, polio, measles, and tetanus; between them, these four categories still cause nearly eight million deaths a year. Other important causes of death are malaria and HIV/AIDS (for which treatment is still far from perfect), deaths at or near birth (perinatal deaths), deaths of mothers associated with childbirth, and deaths from nutritional inadequacies, of which the two most important are deaths from protein or energy insufficiency (not having enough to eat) and deaths from anemia (which comes from a diet that does not supply enough iron, often associated with vegetarianism). Apart from pneumonia, which causes 350 thousand deaths a year among the elderly in rich countries, essentially *no one* dies of any of these causes in rich countries, where better public health measures have greatly reduced the risk of children dying from diarrhea, pneumonia, and tuberculosis. Malaria is not a risk in rich countries, though it was in some countries until shortly after World War II; in poor countries, it mainly causes death among children. Antiretroviral drugs and changes in sexual behavior have greatly reduced deaths from HIV/AIDS. Near-universal immunization of children has largely eliminated the "childhood disease" category, and ante- and postnatal care have reduced perinatal and maternal mortality to very low levels. Few people in rich countries die from lack of food, and while anemia is not unknown, there are no large populations in the rich world that lack vital micronutrients such as iron.

So we have a puzzle. Why should children die in poor countries when they would not die if they had been born in rich countries? What is it that prevents the knowledge that is freely available and effective in the rich world from saving the lives of millions of people who die in the poor world? The most obvious candidate is poverty. Indeed the very classification I have adopted, between low- and high-income countries, suggests that income is what matters. Just as in the historical context, we think of diarrhea, respiratory disease, tuberculosis, and undernutrition as "diseases of poverty," as we think of cancer, heart disease, and stroke as "diseases of affluence." As was the case in the eighteenth and nineteenth centuries, income certainly must play a role; people who have money typically can get as much food as they need, and economic growth helps provide the funds that are needed for vector control, for sanitation and water treatment, and for clinics and hospitals. Even so, the poverty and income story is at best incomplete, and focusing too much on income may mislead us about both *what* needs to be done and *who* should do it.

As always, much can be learned from looking at what happened in China and India. The World Bank no longer counts them as low-income countries, but as lower-middle- (India) and upper-middle- (China) income countries. Both have grown rapidly in recent years, yet they were among the poorest countries in the world in the 1950s. More than a third of the world's population lives in one or the other, so that understanding what happened there is important by any measure. Figure 2 looks at economic growth and infant mortality in the two countries over the past fifty-five years. National income, or more precisely GDP per capita, is plotted on the right-hand vertical axis; once again I have used a log scale, on which a constant rate of growth would show up as a straight line. In fact, for both countries, growth has been *accelerating* over time, particularly—and spectacularly—for China. For India too, after forty years of anemic economic growth, there was acceleration after 1990, particularly at the very end of the period. Both countries instituted economic reforms that are credited with raising growth rates, China after 1970, when farm prices were raised and farmers were encouraged to grow and sell more, and India after 1990, when many of the old rules and regulations of the "license Raj" were scrapped.

Infant mortality rates have fallen as China and India have become richer. The patterns are very similar for child mortality (the 0–4 group), so I do not show them here. The decline in China was halted by the famine, during which as many as a third of the birth cohort died (the figure shows five-year averages, so the effect is much smaller), but the famine aside, the general pattern is of rapid decline until about 1970, followed by much slower decline after 1970. This is precisely the opposite of what we would expect if the fall in infant deaths had been driven by economic growth, which would be the case if the death of babies were a direct consequence of poverty. What happened in China is no mystery. When the authorities decided to focus on growth, resources were switched to making money and away from everything else, including public health and health care. Even the people who were responsible for keeping mosquitoes under control were turned into farmers to join the dash for growth. In the early years, the Communist Party paid a great deal of attention to public health—*Away with All Pests* is the memorable title of an account of a British doctor working in China in the 1950s and 1960s[4]—but that focus was lost after the reforms. None of this means that the reforms were bad; the economic growth after the reforms raised millions of people out of poverty and gave them a better life. What it does show is that the growth does not bring any *automatic* improvement in the health component of wellbeing. In China, it was policy that mattered: in effect, the authorities decided to trade off one aspect of wellbeing for another.

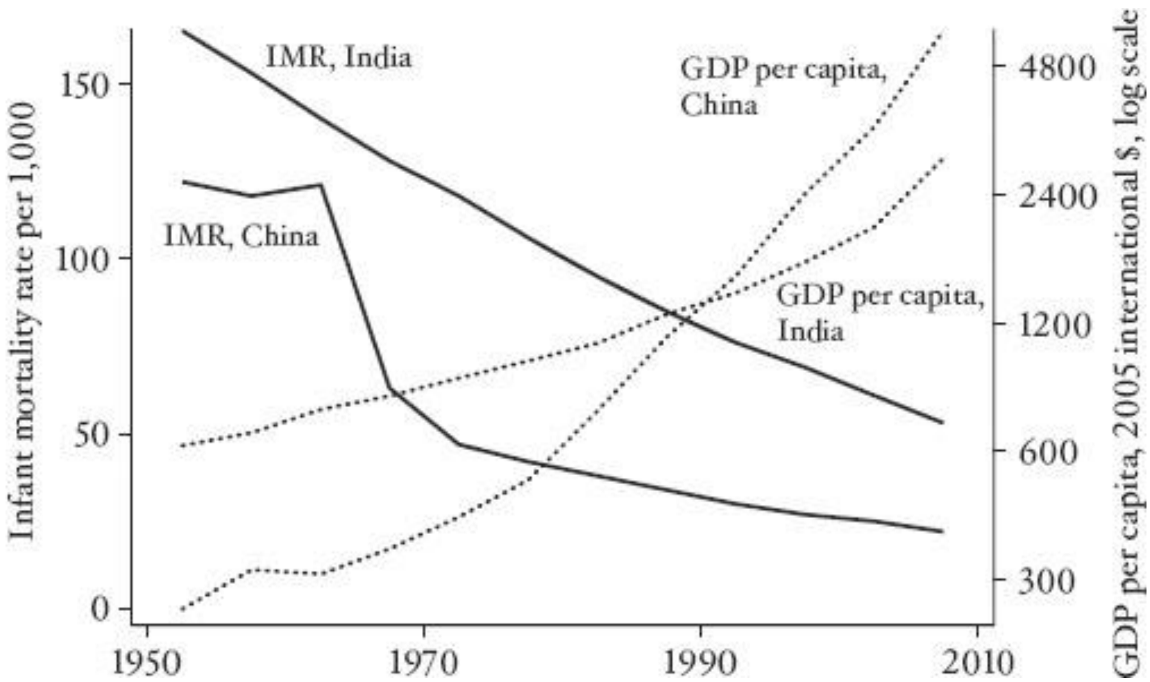

FIGURE 2 Infant mortality and economic growth in China and India.

In India, as always, events were slower and less spectacular. Growth was slower than in China, and the uptick after the reforms less pronounced; India's per capita income used to be higher than China's, but by the early 2000s was less than half of China's. (As we shall see in Part II, these comparisons are subject to *lots* of uncertainty.) Yet India's decline in infant mortality has been remarkably steady—not at all responsive to changes in the rate of growth—and the absolute decline, from 165 out of every 1,000 babies dying in the early 1950s to 53 in 2005–10, is actually *larger* in absolute numbers than the decline in China, from 122 to 22. While it is still more dangerous to be born in India than in China, India's health performance is not obviously inferior to China's, in spite of the very large differences in economic growth. India's success was also achieved without the coercion and loss of freedom associated with the Chinese one-child policy; indeed, as noted by the economists Jean Drèze and Amartya Sen, regions in South India are now doing substantially better than China.[5]

China and India are "only" two countries, and there is no reason why what is true there will also be true elsewhere, so that economic growth may still be the key to health improvements in Africa, or in countries that are much poorer than China and India are today. Yet there is very little evidence that countries that grow more rapidly have had faster declines in infant or child mortality. Figure 3 shows how little relationship there is between how quickly infant mortality has declined and how fast the economy has grown. In order to give the growth story a fair tryout, I look here only at longer-term changes. Rapid growth over a year or two might not do much to bring about the improvements upon which child health depends; for example, a boom in the price of a commodity export might bring in a lot of money for a few people or for the government, but it would have little effect on general prosperity. However, if growth persists for a few decades, its effects should surely show up—if they are really there. The availability of data limits what can be done, but the figure shows growth and mortality decline over spans that are always at least fifteen years long—on average forty-two years long—beginning in some cases as early as 1950 and ending after 2005. The vertical axis shows the annual *decline* in the infant mortality rate, so that bigger is better. Since the infant mortality rate is measured in deaths per thousand, a number like 2 (for India, for example) means that over the years for which I have data (fifty-five years), India's infant mortality rate has fallen by 2 times 55, or 110 deaths per 1,000 births. I have included the rich countries in the picture, but, since they already had low rates of infant mortality, they had small declines over the period, and all cluster at the bottom near the center, so that excluding them would not have made much of a difference to the pattern.

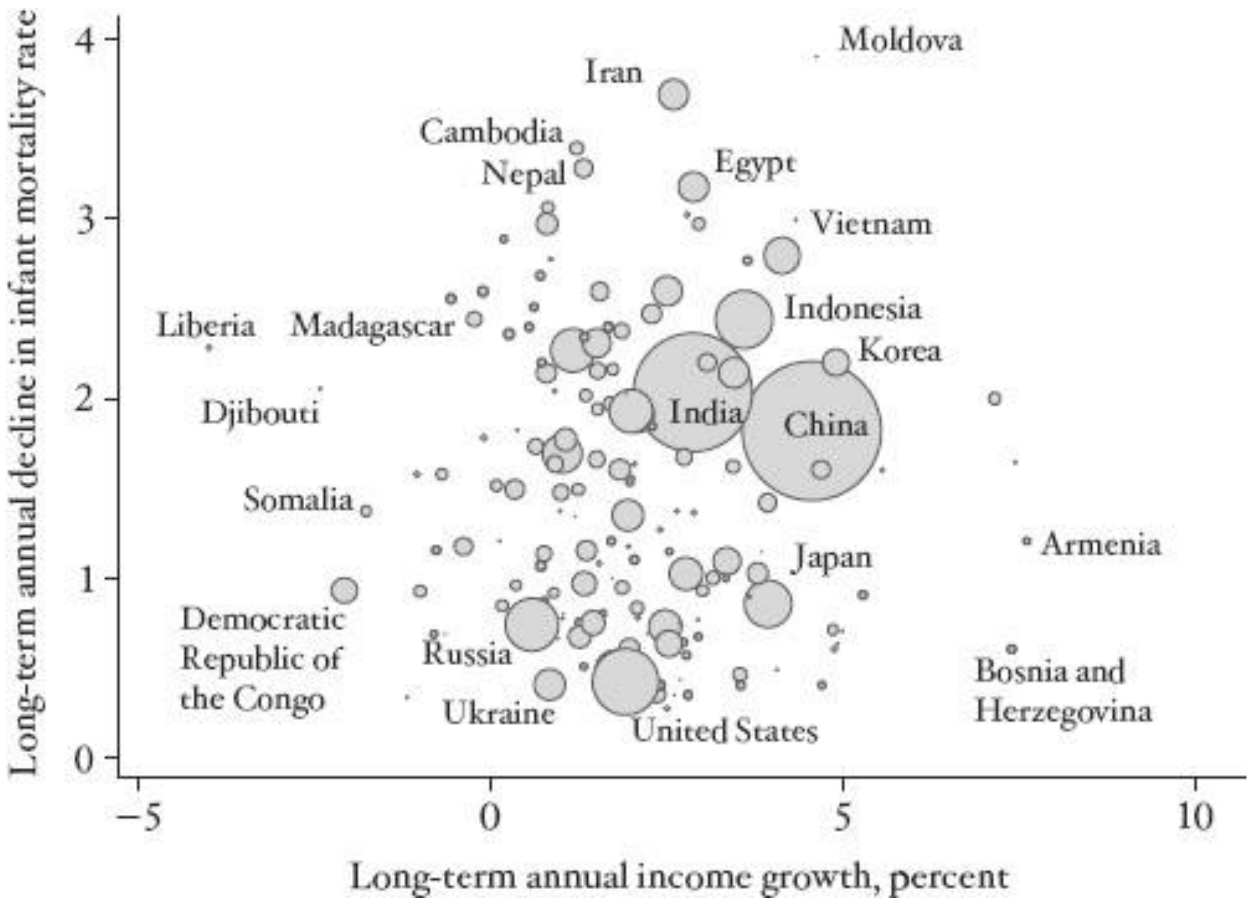

FIGURE 3 Infant mortality and economic growth around the world since 1950.

The figure gives the impression of a positive relationship, but that is because I have followed my usual practice of plotting circles whose area is proportional to population. In this case, there are three big countries, China, India, and Indonesia, that have grown relatively quickly and that have had faster than average rates of mortality decline. However, to check the idea that it is growth that reduces mortality, we should not take population size into account. The question that we are asking is, “Do faster-growing countries have faster rates of decline of infant mortality?” In this respect each country is a separate experiment, and there is no reason to treat different experiments differently. When we look at the picture in that way, and give each country the same weight, there is no relationship at all. At least in the historical record, faster-growing countries did not improve their infant mortality rates at faster rates. The picture shows many examples. Haiti, whose economy actually shrank from 1960 to 2009, had a very respectable rate of infant mortality decline, faster than the rates in China or India. For the sixteen economies that got smaller, the average annual rate of mortality decline was 1.5 per year, slightly better than the rate for all 177 countries in the picture. It is certainly possible for infant mortality to fall even when there is no economic growth at all.

That there should be *no* relationship at all between growth and saving lives is surprising. We know from the historical evidence that other things—like disease control—are as or more important but, even so, it is hard to believe that money does not help at all. And indeed, there is reason to think that Figure 3 may be misleading, because it ignores feedback from the decline in income mortality to the rate of economic growth. When children who would otherwise have died are saved, the population grows, and this may cause income per head to fall, or at least to grow less rapidly than it would have without the life-saving innovations. Eventually, these newly saved children will grow to be productive adults, and there is no reason to suppose, nor any evidence, that larger populations are inevitably poorer populations. Even so, in the first years of lower child mortality, the newly saved people are children, whose contribution to the economy mostly lies ahead of them, so that for a while lower child mortality might reduce each person's share of national income. This effect will work in a direction opposite to any effect of higher per capita income on child mortality and may even cancel it out, giving the lack of correlation in Figure 3.

Yet the evidence does not support this line of argument. It is true that the countries whose infant mortality rates fell most rapidly are also the countries whose populations rose most rapidly. Rich countries, whose infant mortality rates were already low, saw little decline in infant mortality and experienced low population growth. Poor countries saw much more rapid declines in infant mortality, and their population growth was more rapid. But *within* the poor countries, or within Africa, Asia, and Latin America, there is no relation at all between the decline in infant mortality and the rate of population growth, either because other factors were important or because over forty years fertility rates had time to adjust. As we can see in Figure 3, there is no relation between growth and mortality decline, even in the poor countries, and this absence cannot be explained by any obscuring effect of mortality decline on population growth.

If poverty is not the reason why so many children die in poor countries, and if economic growth does not automatically eliminate those deaths, why do they continue, even when most of them are preventable given current medical and scientific knowledge?

It is helpful to turn again to the causes of death listed in Table 1, and to think about how each might be dealt with, because different causes of death call for different solutions. For tuberculosis, malaria, diarrhea, and lower respiratory infections, the environment would need to be different. There would need to be better pest control, better water, and better sanitation, all of which require collective action, organized by central or local government. What might be called the physician-patient health-care system cannot do much about these problems. They are problems of public health, not private health care, even though health care can sometimes alleviate the consequences. Better living standards must surely help too, although, as we have seen from the data, this does not seem to be enough by itself.

Deaths from childhood diseases, from perinatal and maternal conditions, and from hunger could all be prevented by better ante- and postnatal care: giving a mother advice before and after the birth of her child, having health facilities available to deal with emergencies and complications, and having clinics and nurses that monitor young children to check that their immunizations are up to date, to ensure that they are growing as they should, and to advise parents. Children are particularly at risk in poor countries after weaning, when they switch from a relatively rich, complete, and safe diet—breast milk—to a diet that may be insufficient, unvaried, and unsafe. Educated mothers can do a lot by themselves, but doctors, nurses, and clinics can help children and their mothers get through this risky time. For these causes of death, therefore, the physician-patient health-care system is important. Yet many countries spend very little on their health-care systems, and it is close to impossible for a health service to do much good on the $100 per person that is typical for sub-Saharan Africa, a figure that includes private as well as public expenditures. For example, the World Bank calculates for 2010 that, in 2005 price-adjusted dollars, Zambia spends $90 a head, Senegal $108, Nigeria $124, and Mozambique only $49. In comparison, Britain spends $3,470 and the United States $8,362.

Why do the governments of poor countries spend so little when their citizens are in such poor health? Why do citizens in need not turn to private health care when the government is missing in action? And what about the foreign assistance that has been so important in improving some dimensions of international health?

Unfortunately, governments do not always act to improve the health or wellbeing of their citizens. Even in democracies, politicians and governments have a good deal of leeway to pursue their own ends, and there are often sharp political disagreements about what needs to be done to improve health, even when there is agreement on the need to do so. But many countries around the world are not democratic, and more broadly, many governments are not bound to act in the interest of their populations, whether by circumstance—for example the need to persuade citizens to let them raise revenue—or by effective constitutional rules or constraints. This is clearly true in dictatorial or military regimes, or in countries where repressive governments use the armed forces or secret police to control the population. In other cases, governments are well funded by the sale of natural resources—minerals and oil are notorious in this regard—so they have no need to collect revenue from the population. Since he who pays the piper usually calls the tune, governments can use such revenues to maintain a system of cronies and patronage that has little interest in popular health or wellbeing. In extreme cases, particularly in Africa, foreign aid has been significant enough to act in this way too, providing governments with resources but undermining their incentives to spend them in the right way. Even with the best will in the world, it has been difficult for donors to stop this from happening, a topic on which I will have more to say in the last chapter.

Governments do not bear all of the blame. In some places, people do not seem to understand that their health could be better—another place where education might help—or that the government might have the tools to help it improve. In Africa, the Gallup World Poll regularly asks people on what issues their governments should focus. Health concerns are not high on the list, and they appear long after anything to do with poverty reduction or providing jobs; governments that emphasize job creation, even useless jobs in a bloated civil service, may actually be doing what their constituents prefer. In our work in the Udaipur district of Rajasthan, we found that people knew they were very poor, but even though they suffered from a wide range of preventable sicknesses—what the economist and activist Jean Drèze calls "an ocean of sickness"—they thought their health was just fine. It is easy to tell that there are many people richer than you, but much harder to see that they have better health, or that their children are less likely to die; such things are not publicly visible in the way of wealth, housing, or consumer goods.

In Africa, where men and microbes coevolved, the fact that they are both still around is another way of saying that sickness has been man's companion throughout African history. More broadly, and as we have seen in Chapter 2, the escape from sickness and early death happened only recently *anywhere* in the world, and many people may still not understand that such an escape is possible, or that good health care might be a route to freedom. The Gallup World Poll regularly finds that the fraction of people who are satisfied with their health is much the same in poor countries as in rich countries, in spite of huge differences in objective health conditions. There are many countries in the world where people have great confidence in their health-care and medical system, in spite of poor outcomes and low spending. Americans, by contrast, have very low confidence in their health-care system, in spite of all the money that they spend; in one study, the United States ranked 88th out of 120 countries, worse than Cuba, India, and Vietnam and only three places ahead of Sierra Leone.[6]

A great scandal of government health care in many countries is that medical workers—nurses and doctors—are frequently absent from work. In Rajasthan, only about half of the small clinics were open at all when we made random checks, and while the larger ones were open, many of the health workers were not there. The World Bank has carried out surveys on absenteeism, and it turns out that in many countries—although certainly not all—absenteeism is a huge problem in both health care and education.[7] In some cases, these workers are not paid very much. It is as if there were an implicit contract between the workers and their employers; the government pretends to pay them, and they pretend to show up for work. But low wages are not always the reason. When people expect little of their health service, it is easy for absenteeism to flourish. In Rajasthan, it was hard to get people even to admit that a particular nurse had not shown up for weeks, and for many, this level of service is what they expect of the public system. But not everywhere. The Indian state of Kerala is famous for its grass-roots political activism, and for the robust protests that follow the failure of a clinic to be open. In Kerala, absenteeism is rare, and people expect their clinics to serve them. If we knew how to move Rajasthani attitudes closer to Keralan attitudes, a large part of the problem would be solved.

Private physicians often do a flourishing trade in poor countries, and their services often help make up for the deficiencies of state-provided (or not provided) health care. But the private sector has problems of its own. In particular, knowing what you need when you are sick is a problem for anyone who is not a trained physician. Buying health care is not like buying food when you are hungry; it is more like taking your car to the repair shop. The people who are better informed are the very people who are providing the care, and they have incentives and interests of their own. In the private sector, providers make more money if they provide more care or more profitable care; they also have incentives to give people what people think they want, whether or not they actually need it. In India, private practitioners routinely give people the antibiotics that they demand, often by injection, leaving them as satisfied consumers and feeling (temporarily) better. Intravenous drips are another favored item, and they are heavily advertised by health-care providers in India, just as complete body scans or PSA tests for prostate cancer are relentlessly marketed in the United States. Public doctors in public clinics and hospitals in India typically will not give antibiotic shots or intravenous drips on demand—a good thing—but they also do not have time to carry out tests to find out what a patient might actually need—not such a good thing. So the choice between a public and a private doctor is a matter of chance, though you are likely to *feel* better treated—at least in the short run—when you visit a private doctor.

All of this would be less of a problem if public-sector health care were trustworthy, or if private-sector health care were properly regulated. The problem in many countries is that neither condition applies. Indeed, even in the world's richest countries, the provision and regulation of health care is one of the most difficult, contentious, and politically charged functions of government. Most of the private "physicians" visited by the people we talked to in Rajasthan were not qualified doctors but quacks of one kind or another—what in Rajasthan are slightingly referred to as "Bengali doctors." Several "doctors" had not even graduated from high school. The lack of government capacity lies behind the failures of *both* private and public health care. The government is capable neither of delivering health care itself nor of providing the regulation, licensing, and policing that is required for an effective and safe private health-care system.

Money is a problem too, and it is probably true that India (and many countries in Africa) could not run a better health-care system without spending a great deal more than is currently spent. However, it is also easy to imagine a much more expensive system that is no better, in which absentee doctors get paid even more for not showing up for work. Without an educated population and without government capacity—an effective administrative structure, cadres of educated bureaucrats, a statistical system, and a well-defined and enforced legal framework—it is difficult or impossible for countries to provide a proper health-care system.

FOUR

Health in the Modern World

SINCE WORLD WAR II, people in poor countries have begun to see the health benefits that people in rich countries have long enjoyed. The germ theory of disease had made possible a great reduction in the burden of infectious disease, but the science and science-based policies took more than a century to spread from their origins to the rest of the globe. If this had been the whole story, the late adopters would have eventually caught up with the pioneers, and the history of global health would have been the story of the gradual elimination of the inequalities in international health that first appeared in the eighteenth century. But there were further escapes to be made, even in the pioneer countries, and the length of life continued to increase, even in the countries that had led the way and even after deaths of infants and children had become rare. Now it was the turn of the middle-aged and the elderly.

This chapter tells the story of how those further escapes came about, and what the future might hold for longevity in the rich world. It is also about the implications for health of a highly interconnected world in which it makes less and less sense to talk about the rich world and the poor world. With faster and cheaper transportation and communication, health innovations in one country have almost instantaneous implications for health in the rest of the world; the germ theory may have taken a century to spread, but modern discoveries travel much more rapidly. New diseases, like new treatments, also ride the global highways. In this age of globalization, international inequalities in longevity have been shrinking. However, longevity is not the only aspect of health that is important, and it is much less clear that international *health* inequalities are getting smaller; they certainly should not be thought of as relics ready to go quietly into the dustbin of history. Health is not only about living and dying, but about how healthy people are while they are alive. One measure of “living” health, which provides both an antidote and a complement to life expectancy, is human height—a sensitive indicator of the burden of undernutrition and disease, especially among children. We will see that most—but not all—of the people of the world are growing taller. Yet progress is slow; at current rates, it will take two hundred years for Indian men to grow as tall as Englishmen are now. And that is not the worst news; it will take nearly five hundred years for Indian women to catch up with English women.

The Elderly Can Escape Too: Life and Death in the Rich World

Even in the rich countries, the health improvements that came from the germ theory were far from complete by 1945; the infant mortality rate in Scotland in that year was as high as it is in India today. Yet, after World War II, increases in longevity in the pioneer countries came increasingly to depend on reductions in mortality among the middle-aged and elderly, and less on reductions in mortality among infants and children. Today, the leading causes of death are no longer tuberculosis, diarrhea, and respiratory infections, but heart disease, stroke, and cancer. Yet life expectancy is still increasing (albeit more slowly than it did before 1950), driven upwards not by cleaner water and more complete vaccinations but by medical advances and changes in behavior.

By 1950, the rich countries of the world had done most of the work of escaping from childhood infectious disease, and by 2000, the job was essentially complete. As I write, in 2013, around 95 percent of all newborns in rich countries can expect to reach their fiftieth birthday. In consequence, further increases in longevity now depend on what happens to the middle-aged and elderly. Here too there has been much progress in the past fifty years.

Figure 1 shows what happened to life expectancy at age 50 in fourteen of the world’s rich countries. Life expectancy at 50 is defined as the number of years that someone on his or her fiftieth birthday can Japan expect to live, so if life expectancy at 50 is 25 years, the 50-year-old can expect to live to age 75; as with life expectancy at birth, the calculation assumes that mortality rates will remain constant. The figure shows the average for men and women; as always women do better, but here I simply want to show the rate of progress for everyone, rather than looking at differences between the sexes. Even in 1950, the 50-year-olds in all of these fourteen countries could expect to live at least a couple of years beyond the biblical threescore and ten; this was true even in Japan, which was then the worst-performing country in the group. There was substantial inequality across countries in 1950, from 27.0 years in Norway to 22.8 in Finland and 22.6 in Japan. In the 1950s and 1960s, progress was different in different countries, but after 1970, the rate of increase in longevity speeded up. It was also much more closely synchronized across countries. Whatever was making people live longer seemed to be working in much the same way everywhere. Between 1970 and 1990, life expectancy at 50 in these countries rose by nearly three years. Progress continued after 1990, but now there were more differences across countries; some, like Japan, did extraordinarily well, while others, like the United States and Denmark, lagged behind.

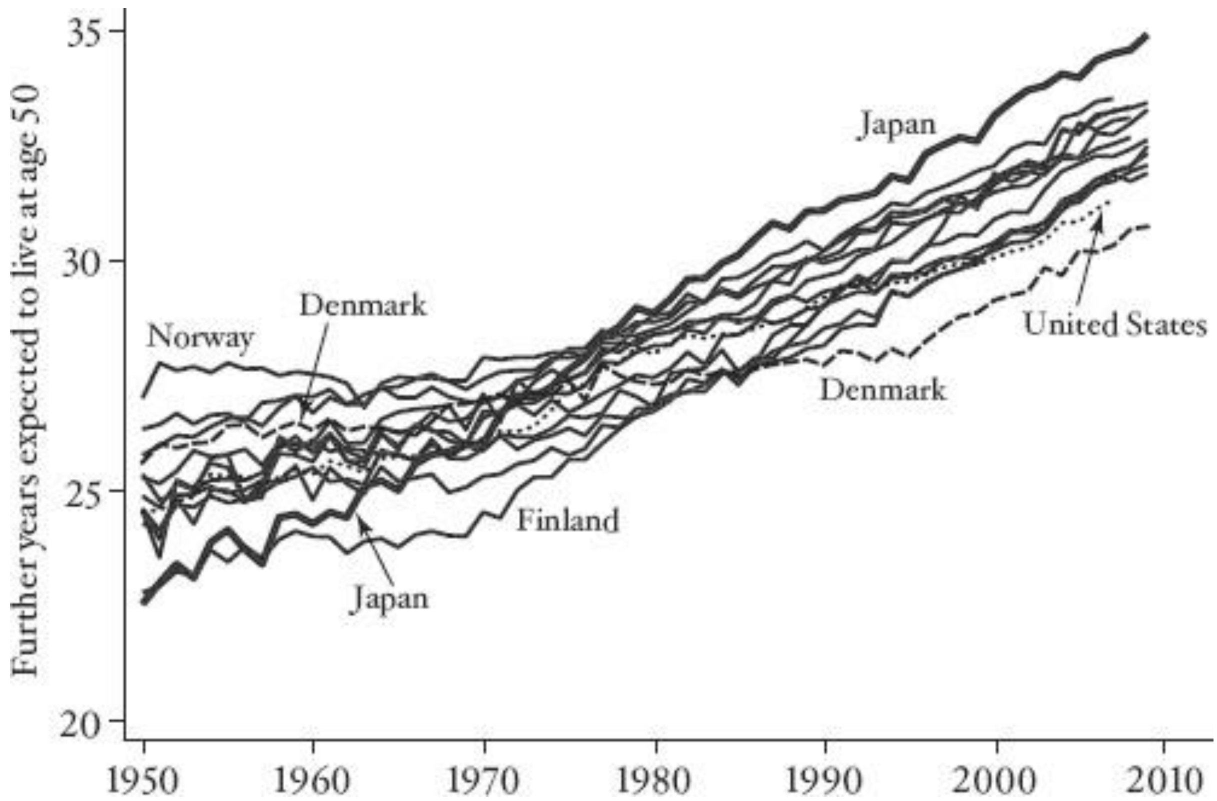

FIGURE 1 Life expectancy at 50 in wealthy countries (men and women together).

The main message of the figure is that the middle-aged and elderly everywhere saw big reductions in mortality rates after 1950. As we saw in Chapter 2, nothing like this happened prior to 1950, when the improvements were mostly among children, with less of a gain in life expectancy at higher ages. The second message is that some countries have done better than others. Japan, which was last in 1950, is now first. Denmark, which was among the early leaders, is now last, and the United States, which started out in the middle of the pack, is now second from the bottom.

What made this happen? One reason transcends the specifics of diseases and their treatment. People do not want to die, and they will devote great resources—both their own and those of their governments—to trying to escape death. When large fractions of children die before they become adults, doing something about child mortality is the first priority for parents and for society as a whole. But as people live longer, the "next" disease becomes the most important one, and the "next" disease typically means the next major killer that afflicts people at an older age than did the "last" one. Having slain the first monster in the labyrinth, the next priority is the monster that lurks behind it, whose presence becomes much more important once we have figured out how to dispose of the first one.

With child mortality and infectious disease largely behind us in the 1960s and 1970s, the next monsters were the chronic diseases that killed people in middle age: heart disease, stroke, and cancer. *Chronic* in this context applies to diseases that last for a while—conventionally more than three months—and is the opposite of *acute*, referring to diseases that threaten to carry you off quickly, as happens with many infectious diseases. (Perhaps more accurate would be the descriptors *noncommunicable* and *infectious*.)

As we shall see, there has been progress against all three of the major chronic diseases, particularly heart disease and stroke, which both fall into the category of cardiovascular disease. At least some of this progress came from people being prepared to spend large sums of money, partly on treatment, but more importantly on the research and development that unraveled the basic mechanisms of the diseases and thus allowed the design of better treatments. As cancer and cardiovascular disease recede in importance—as we can reasonably hope they will—new urgency will be focused on disorders such as Alzheimer's disease, a condition that was much less of a priority in 1950, let alone 1850, because so few people lived long enough to suffer from it. Just as in the nineteenth century, new diseases demand new cures and offer new opportunities to discover them. Today, as death itself ages, the challenging diseases are those that afflict older and older people.

Cigarette smoking is one key to understanding recent mortality trends in high-income countries.[1] The patterns are not the same everywhere, but the spread of smoking in the first half of the century happened everywhere, with subsequent declines in many, if not all, countries. At first, women were much less likely to smoke than men, so that women became smokers later and, in countries where smoking is now declining, have been slower to quit than men. Smoking brings benefits to people in straightforward enjoyment, and it was a cheap and sociable pleasure for poor and rich alike. For many poor people, it was an easily available and affordable activity that provided a temporary escape from busy and often difficult lives. But it also brought disease and death. Lung cancer is most strongly associated with cigarette smoking because very few people who die of lung cancer did not smoke, although not everyone who smokes gets lung cancer. Deaths from lung cancer typically lag behind smoking trends by about thirty years, so that the mortality from smoking continues long after the behavior has changed. But cigarettes likely kill more people through cardiovascular disease than lung cancer, and there are other unpleasant consequences, such as respiratory diseases. The most important among these is chronic obstructive pulmonary disease, which includes both bronchitis and emphysema; the disease makes it difficult to breathe and is a major cause of death.

In the United States, the publication in 1964 of the surgeon general's *Report on the Health Consequences of Smoking* (for men!) is often seen as the hinge moment for behavioral change; many older Americans will say that they smoked until the report came out, but then either quit, or at least resolved to do so, immediately thereafter. There was no better example than that of the surgeon general himself, Dr. Luther Terry. In an attempt to minimize public attention, the press conference for the release of the report was scheduled for a Saturday morning in Washington, D.C., and as he traveled to the conference in his limousine, Dr. Terry was smoking. To his intense irritation—"that is none of their business"—an aide warned him that the first question would be whether or not he himself was a smoker. Indeed it was, and Terry replied with an unhesitating "No." "For how long?" was the follow-up. The answer: "Twenty minutes." Millions of Americans followed the surgeon general's example in the years that followed. Sales of cigarettes peaked in the early 1960s at around eleven per day for each adult, when about 40 percent of the population smoked, each consuming more than a pack a day.

That the surgeon general's report by itself changed everything can reasonably be doubted. There had been many earlier reports on the health consequences of smoking—indeed, my mother was ordered by her doctor in Edinburgh in 1945 to quit smoking during her pregnancy, which may be why I am writing this book—and even in the United States, the 1964 peak was largely coincidental. Smoking among men had been declining well before 1964, and women's smoking had been rising for some time; it was only the sum of the two that peaked in 1964.

Knowledge about the harmful effects of smoking is now widespread, at least in rich countries, so one might think that smoking should have declined everywhere. Yet there remain significant differences across countries and between men and women. Incomes and the local cost of cigarettes vary from one country to another, and different countries have different attitudes about health warnings and restrictions on smoking in public places. None of those factors does much to explain the differences between men and women. In some countries, it was socially disreputable for women to smoke—in Scotland in the 1950s women who smoked on the street were regarded (at least by my mother) as little more than prostitutes—and the right to smoke became associated with movements for equal rights for women. In the United States, as in Britain, Ireland, and Australia, women's smoking caught up with or even exceeded that of men, although today the prevalence of smoking is falling in both sexes. In Japan, the rate of smoking among men has been extraordinarily high (close to 80 percent in the 1950s), though it is declining now; very few Japanese women have ever smoked. In continental Europe, smoking is also generally declining, but there are many exceptions, especially among women. As someone once joked, the surgeon general's report was not translated into "foreign" languages.[2]

There is a parallel between the spread of smoking and the spread of the germ theory of disease less than a century before. Cigarettes are, or were, part and parcel of the way people lived, and they are, or were, an important source of pleasure. The knowledge that cigarettes are harmful makes people less likely to smoke, but there are offsetting considerations—not to mention a habit that is difficult to break. Knowledge of the germ theory needed to be embodied in day-to-day housekeeping and hygiene, and it too involved habits and ways of living that were difficult and sometimes costly to change. In both cases, gender roles were important. Women were primarily responsible for the housekeeping and childrearing tasks that were important for implementing measures against the spread of germs, and in many families, women became the household "germ police."[3] In the case of cigarettes, smoking was linked first to the oppression and later to the liberation of women. It is also important to keep in mind that cigarettes are *not* analogous to cholera bacteria or the smallpox virus, in spite of the current demonization of tobacco and the frequent use of the terms *plague* or *epidemic* to describe smoking. Certainly smoking is harmful to health, but it also brings benefits, something no one ever claimed for the bubonic plague or, for that matter, for breast cancer. It is not a mark of insanity if a person decides that the pleasures of smoking more than compensate for the health consequences. Many localities in the United States are currently raising substantial sums of money from the predominantly poorer people who choose to smoke; these funds are largely used to offset property taxes for better-off people. It is far from clear that any overriding public health interest justifies this taxing of the poor to benefit the rich.

The rise and fall of cigarette smoking is echoed in the rise and fall of deaths from lung cancer in Figure 2.[4] The graphs show the mortality rate for people aged 50–69 from lung cancer since 1950 for Australia, Canada, New Zealand, the United States, and the countries of northwestern Europe. The United States is shown as the heavy line in both graphs. In the graph for men, we see an explosion of mortality, peaking around 1990, about two to three decades after the peak in smoking, and then falling back. On the right, because women took up smoking much later, the fall is confined to only a few countries, and the graph looks like the open jaws of a crocodile. Among women, the explosion in smoking is still going on, although for a few countries, including the United States, lung cancer mortality has begun to fall. Women never smoked as much as men, so their mortality rates are lower, in line with their smoking rates in earlier years, with the countries in which women smoked experiencing higher death rates. Note finally that, although lung cancer is an important cause of death, only a small fraction of the 40 percent who used to smoke actually died (or will die) of lung cancer; the average annual mortality rate in the United States in the worst years was just over 200 per 100,000 people, or a fifth of 1 percent.

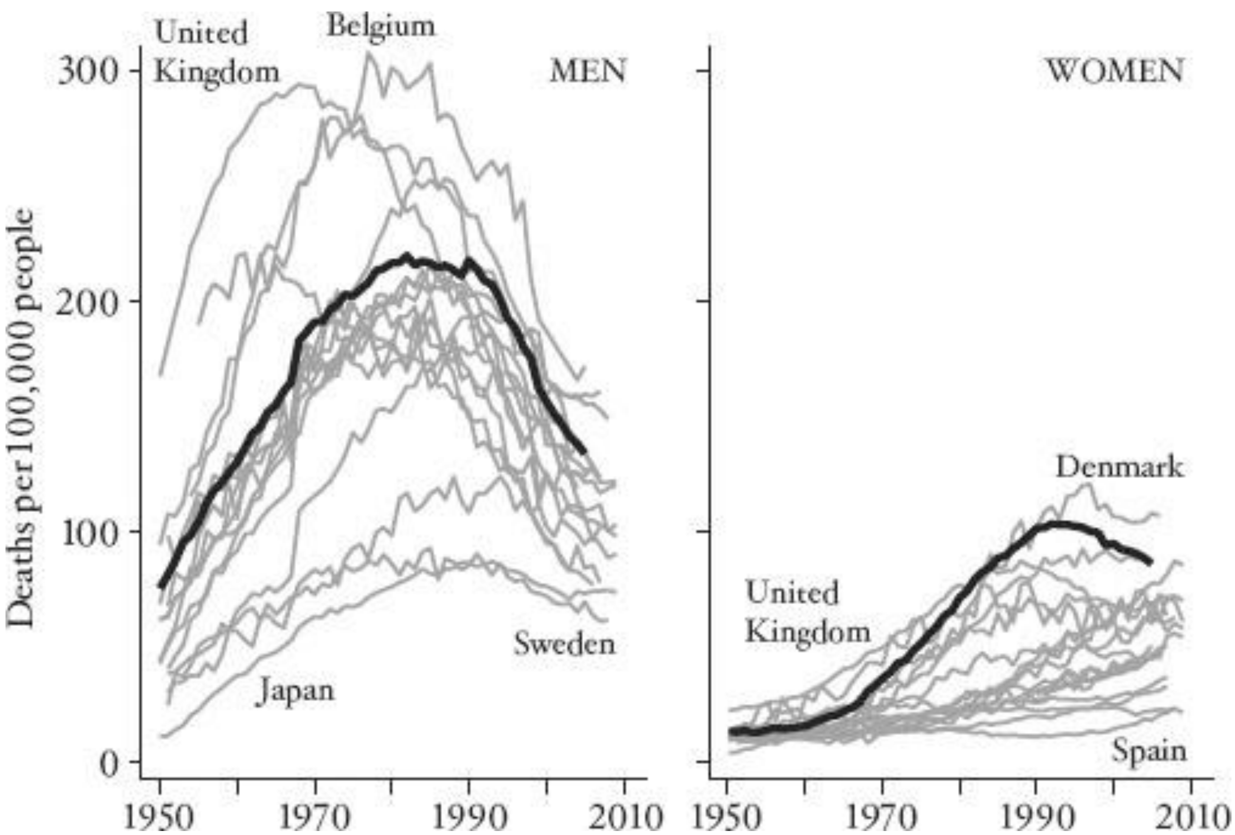

FIGURE 2 Mortality from lung cancer (heavy line is United States).

Although smokers are ten to twenty times more likely to die of lung cancer than nonsmokers, the vast majority of smokers do not die of the disease; the Memorial Sloan-Kettering Cancer Center has an online calculator that estimates the risks.[5] For example, a 50-year-old man who has smoked a pack a day for thirty years has a 1 percent chance of developing lung cancer if he quits now and a 2 percent chance if he does not. Before anyone takes too much comfort from this information, it should be remembered that lung cancer is neither the only nor the most prevalent risk from smoking.

Cigarette smoking is the main reason that women's life expectancy has been rising less rapidly than men's in recent years, not only in the United States but also in a number of other countries where women started smoking early, including Britain, Denmark, and the Netherlands. American women are paying quite a price for the tobacco companies' successful attempt in the 1960s and 1970s to link women's liberation to cigarette smoking. American smoking prevalence is the single biggest reason why life expectancy at 50 has been growing less rapidly in the United States than in a number of other rich countries, such as France and Japan. Recent calculations estimate that without smoking, life expectancy at 50 in the United States would be 2.5 years longer than it is.[6]

Even more important than the decline in lung cancer has been the decline in deaths from cardiovascular disease, a term that covers diseases of the heart and veins, including stroke, atherosclerosis (the accumulation of plaque that blocks arteries), coronary artery disease, heart attack, congestive heart failure, and angina. The reduction in smoking among men has helped relieve this burden too, but there have also been important advances in effective medical treatment, something that has not so far been true for lung cancer.

Figure 3 shows mortality from cardiovascular disease since 1950 for middle-aged to older men between the ages of 55 and 65. In the left panel, I show only the United States and Britain; in the right panel, I show mortality in the same rich countries included in Figure 2. These numbers are *huge*—about *five times* the mortality rate for lung cancer. In the 1950s, between 1 and 1.5 percent of these middle-aged and older men could expect to die in any given year. Cardiovascular disease was then, and remains today, the leading cause of death in United States high-income countries. In the 1950s and 1960s, mortality from cardiovascular disease was higher in the United States than in Britain, rising slowly in Britain and falling slowly in the United States. Among other rich countries, the United States had the highest risk, and there was considerable variation across countries, with Iceland and the Netherlands at the bottom of the graph. Until about 1970, each country went its own way, with no obvious coordination across countries. Whatever the cause of cardiovascular disease, it was—like smoking, which is indeed one of its causes—different in each country.

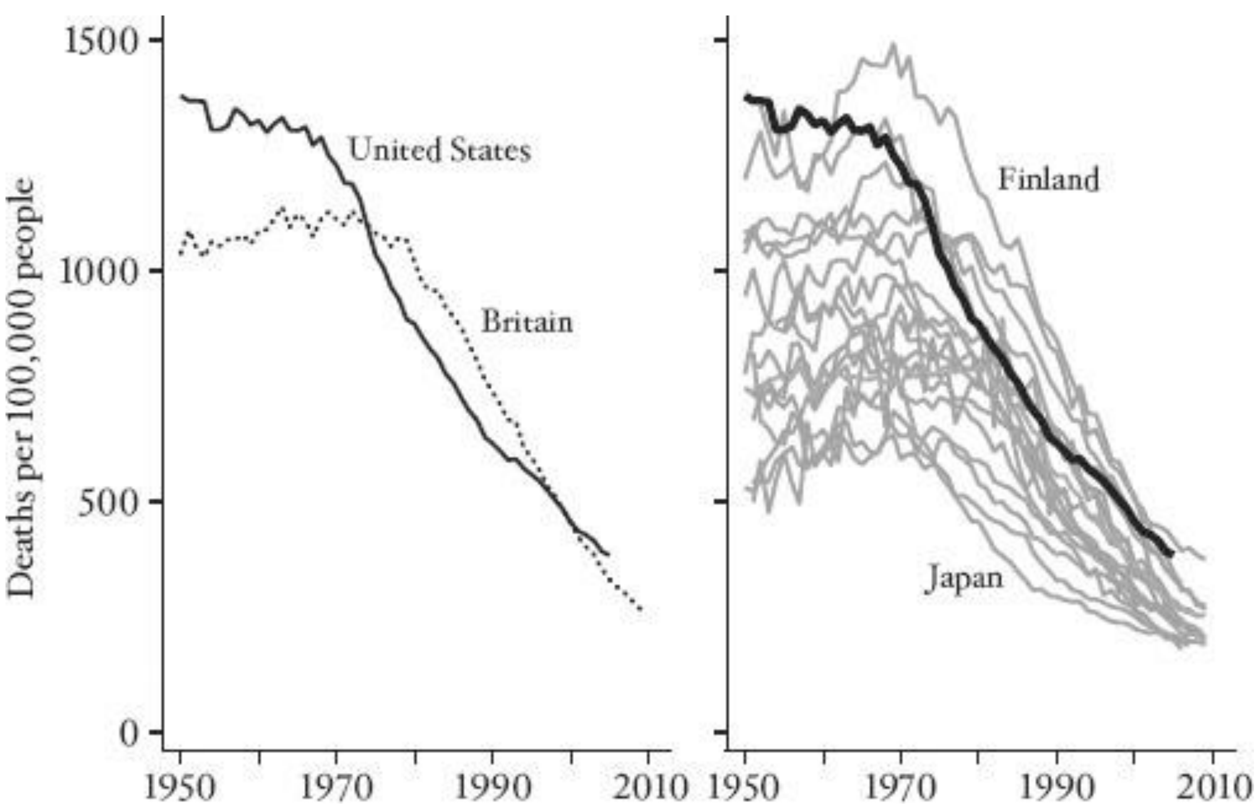

FIGURE 3 Mortality from cardiovascular disease (heavy line in right-hand panel is United States).

Everything changed after 1970. With the United States leading the way, mortality from cardiovascular disease started to fall—later in some countries than in others; Britain, for example, was seven or eight years behind—and there was an internationally synchronized collapse of mortality from cardiovascular disease. Even Finland—once the home of mortality from cardiovascular disease, with annual mortality of 1.5 percent in 1970—rapidly fell into line, so that in the first years of the twenty-first century not only has the mortality rate fallen by a half to two-thirds, but there has also been a convergence of rates across different countries. Nearly all of the diversity of experience in the 1950s has vanished.

What happened? Quitting smoking was part of the explanation, but as we have already seen, behavior varies from country to country even now, and behavioral change is unlikely to be so fast and so coordinated across countries. It is not as if there were some international health authority—and the World Health Organization is hardly a likely candidate—that ordered all its member countries to change at the same time. A much better candidate is a medical innovation, especially one that is both cheap and effective, so that it can quickly move from one country to the next.

One key innovation in managing cardiovascular disease was the discovery that diuretics—cheap pills, sometimes called "water pills" because they increase the frequency of urination—are effective antihypertensives, meaning that they reduce high blood pressure, one of the major risk factors for heart disease. According to the Mayo Clinic, "Diuretics … help rid your body of salt (sodium) and water. They work by making your kidneys put more sodium into your urine. The sodium, in turn, takes water with it from your blood. That decreases the amount of fluid flowing through your blood vessels, which reduces pressure on the walls of your arteries."[7] An important randomized controlled trial from the U.S. Veterans Administration was published in 1970,[8] and thereafter practice changed quickly in the United States.

One of the characteristics of the U.S. health-care system is that innovations tend to be introduced very quickly—not only the good ones like antihypertensives, but also many that are of dubious value. Britain, with its cash-constrained and centrally run National Health Service, tends to be much slower and more cautious about introducing medical innovations—today it has a National Institute of Clinical Excellence, with the splendid acronym NICE, to test new products and new procedures and make recommendations—so even the cheap and effective diuretics took a while to be adopted. The right-hand side of Figure 3 shows that the same happened elsewhere; the United States led the way, and other countries followed after a time that varied with local institutions and health-care systems.

Diuretics were the first antihypertensives and they were followed by a series of others—with names like ACE inhibitors, calcium channel blockers, beta-blockers, and angiotensin antagonists—so that physicians now have a large cast of characters from which to select the one that best suits a specific patient. Cholesterol-reducing drugs—statins—have also contributed to the mortality reduction, by one account by as much as drugs that reduce blood pressure.[9] These preventive measures are designed to reduce the chance that people get sick in the first place, but there have also been innovations in treatment. One important—and also very cheap—treatment is to make sure that people who are brought to a hospital after a heart attack are immediately given aspirin. There are other, higher-tech innovations for treating heart disease—bypass surgery and the like—which are decidedly *not* cheap, and which may also have contributed to the decline in mortality. One clinical trial showed that on average there were mortality reductions for middle-aged people who took a "baby" aspirin every day, but it has subsequently become clear that while such a treatment saves some, it kills (a smaller number of) others—a good example of what is often a sharp conflict between the average and the individual. Even so, the innovations in treatment and prevention have collectively saved millions of lives, greatly reducing mortality from the leading cause of death; leaving millions of middle-aged men who would otherwise have died to continue working, earning, and loving; and making it more likely that they will get to meet and know their grandchildren.

What about women? As is the case for lung cancer, the mortality rates from cardiovascular disease are *much* lower for women, generally half of those for men. But those rates too have been decreasing, by around a half depending on the country, and with a similar degree of international coordination, so that international variations in women's mortality rates from cardiovascular disease today are much lower than was the case in the 1950s. Although they had lower risks to begin with, women have shared with men the benefits of declining risks of dying from heart disease. For women, as for men, cardiovascular disease is the leading cause of death. While breast cancer is (rightly) seen as an important and specific threat to women, fewer women die of breast cancer than die of heart disease.

The innovations that have helped prevent and treat cardiovascular disease are unusual because they did *not* generate cross-country inequality among the relatively rich countries—rather the reverse. Rates of mortality from heart disease are much more similar across countries now than they were half a century ago, so the important innovations that drove the decline did not generate the international inequality in health outcomes that was brought about by the germ theory of disease a century ago. Perhaps because the important innovations were cheap and easily imitated, countries could quickly bring them into their health services. But cheapness does not seem to have helped guarantee complete diffusion *within* countries, and progress against cardiovascular disease has likely caused some widening of health inequalities across income and education groups. The part of the treatment that depends on the individual—in this case regular visits to a physician to have blood pressure taken and cholesterol measured—were adopted more quickly by the more educated, the better off, and those who were already healthier.[10]

Cancer is the second major killer after heart disease. After lung cancer, the most important forms of the disease are breast cancer (almost entirely among women), prostate cancer (entirely among men), and colorectal cancer (which affects both men and women). At least until the 1990s, there was little progress in treating these cancers, and mortality rates did not fall. In spite of a multi-billion-dollar war on cancer in the United States, people went on dying at about the same rates, and the most authoritative reviews concluded that the war was being lost, or at least not won.[11] Throughout this book, I have emphasized that the discovery of new knowledge and the invention of new ways of saving lives are responsive to the need for them. But demand does not *always* create supply, nor do billions of dollars or the declaration of war on a disease necessarily cure that disease—as evidenced by the failure to find a cure for cancer.

Yet, once again, there is evidence that, at last, progress is being made, and that mortality rates for all three kinds of cancer have begun to fall.[12] This decline may have been going on for some time, but, perhaps paradoxically, it might have been disguised by falling rates of mortality from cardiovascular disease. If we get better at dealing with the first monster in the labyrinth, the monster behind it claims more victims, and it will kill more even if it is not as deadly as it once was. People who were saved from heart disease became available to be afflicted by cancer, and if some of the risk factors (perhaps obesity) overlap, then successes in preventing cardiovascular disease should raise mortality from cancer. That this did *not* happen, that the dog did not bark in the night, might then be counted as evidence of progress against cancer. But recent reductions in cancer mortality provide more direct evidence of success. Screening for all three diseases—with mammograms, PSA tests, and colonoscopies—is often given some of the credit, though its role, especially that of mammograms and PSA tests, cannot be very large. For example, with the rise in mammography, there was an enormous increase in early-stage diagnoses, but no sign of the corresponding decline in late-stage diagnoses that should have been the consequence; over the past thirty years breast cancer screening has detected cancer in more than a million women who would never have experienced any symptoms.[13] Improvements in treatment have likely been more important, such as the use of tamoxifen for breast cancer. In his biography of cancer, *The Emperor of All Maladies*, oncologist and historian Siddhartha Mukherjee argues that, after generations of what was essentially trial and error in surgical and chemical treatment, a better scientific understanding of the origins of individual cancers is slowly emerging and is beginning to pay off in new and more effective treatments.[14]

In contrast to many of the most effective new treatments for cardiovascular disease, the new chemical and surgical treatments for cancer are often very expensive, and that expense will limit the speed with which they are transmitted to other countries. Screening itself is not very expensive, but it can cause great subsequent psychic and monetary expense. A prime example is the situation in which screening detects not a disease itself but a risk factor for a disease, such as high blood pressure, high cholesterol, or even a genetic predisposition. Treating those in whom such risk factors are detected—with antihypertensives, statins, or in extreme cases with prophylactic surgery, such as breast removal for women with a genetic risk of breast cancer—will save the lives of a few of those treated, while treating a much larger number of healthy people who would never develop the disease.[15] When screening is effective, it can also introduce inequalities if the more educated and the better informed adopt it first. Even so, there is hope that screening will become more effective over time, that overscreening will be better controlled, and that drugs and procedures will become cheaper as they are more widely prescribed. If so, there is a good hope that cancer will follow cardiovascular disease as one of the success stories of science and medicine. One more bar of the prison of poor health will have been removed, giving people more capabilities to lead better lives for more years.

Many other factors affect mortality rates, though these are typically less clear or more controversial than the ones that I have discussed. One is our old friend, more and better food. Better nutrition is more plausible as a factor driving down mortality rates in the nineteenth century, when hunger was more common than it is today; today we tend to worry about people eating too much, not too little. Even so, it is possible that one of the reasons that mortality rates are falling among the elderly today is improvements in their nutrition seventy years ago, when they were being conceived, born, and nurtured as children. Finland, which had the highest mortality from cardiovascular disease in the 1970s, was one of the poorest countries in the world around the time of World War I, when the 55-year-olds of the 1970s were born.

Another piece of evidence that supports the food argument is a remarkable finding by the demographers Gabriele Doblhammer and James Vaupel.[16] They calculated that in the Northern Hemisphere life expectancy at 50 is half a year higher for people born in October than for people born in April. The pattern is reversed in the Southern Hemisphere, except for those born in the North and who later emigrated to the South; they too show the Northern pattern. One plausible reason for their finding is that, even in now-rich countries, green leafy vegetables, chicken, and eggs used to be readily and cheaply available only in the spring, which meant that nutrition in the womb was better for unborn children whose due date was in the autumn. As might be expected, this effect has become smaller over time, as seasonal differences in food supply have become less pronounced.

Declining mortality is a great boon—we all want to live longer—but it is not the only kind of health improvement. We also want to live better and healthier lives, so we should not focus only on mortality and ignore *morbidity.* People who are physically or mentally disabled, or who suffer from chronic pain or depression, have fewer capabilities to do the things that make life worth living. There have been important improvements here too. One is the development—essentially through trial and error—of joint replacement, particularly hip replacement, which is now a routine procedure that relieves what would otherwise be a lifetime of pain and immobility.[17] Hip replacement is one of those "magic" surgeries that turns a difficult, painful, and limited life into one in which original function is almost completely restored. Similarly, modern cataract surgery restores or even improves vision. Such procedures restore a whole range of capabilities that would otherwise be lost. Pain medication is much better than it was; ibuprofen (available since 1984) provides relief in situations where aspirin does not, and health professionals now understand much better how to allow patients to control their own pain medication in more serious situations. New drugs for the treatment of depression have improved many people's lives. Access to health professionals is important even when they can do nothing, because they can at least reassure people who are concerned about their own or their loved ones' health; and even if they cannot, they can help resolve the uncertainty that is a source of distress on its own.

Physician care and treatment cost money—from either individuals, their insurers, or the state. The United States spends a uniquely high amount on health care—currently about 18 percent of national income—but it is not unique in facing challenges in paying for ever more expensive and, in many cases, *effective* new techniques. In some cases, in order to save money, states place restrictions on access. In one famous story, the British National Health Service in the 1970s severely limited the availability of kidney dialysis, restricting it to those who were deemed young enough to benefit and excluding those in their 50s, who were described as "a bit crumbly" and not worth the cost.[18] In some periods, Britain has also had long waiting lists for hip and knee replacements. In such cases, inadequate provision of health care raises morbidity and mortality rates, and access to kidney dialysis and joint replacement is currently much less restricted in Britain. Yet Britain has not abandoned its attempts to control the introduction of new drugs and procedures. I have already referred to its NICE, which tests medical innovations and issues detailed reports on how well they work and whether they offer value for money. Such an institution is strongly opposed by the pharmaceutical industry and by device manufacturers. At least one pharmaceutical company threatened to withdraw from Britain after an early adverse decision, but Tony Blair, then the British prime minister, held his ground.[19]

There is disagreement among both economists and doctors on how much health care is too much or on the necessity for some form of rationing. Some point to the enormous successes of medicine; they argue that if we put reasonable values on morbidity and mortality reductions—which doctors hate to do, and which is an imprecise and controversial art at best—we need more health care, not less, even in the United States. Spending twice as much money and getting twice as much mortality and morbidity reduction, they argue, would still be a good deal. Some of these calculations make the mistake of crediting *all* mortality reductions to health care—ignoring, for example, the large effects of reductions in cigarette smoking—but even with a more reasonable attribution, a case can be made for spending more, not less. As we get richer, this argument goes, what better way to spend our money than on achieving better and longer lives? And if health care costs more in the United States than in Europe, that is in part because health care is more luxurious in the United States—more private or semiprivate rooms in hospitals, shorter waits for diagnostic tests and screening—which makes sense, given that Americans are on the whole richer than Europeans and can afford such things.

The opposing argument concedes that health care has delivered great benefits but focuses on the waste in the system, which affects the level of expenditure, and on the lack of a NICE-like approval process, which allows new procedures to be introduced whether or not they are beneficial and accelerates the rate of growth of spending. One of the star witnesses in making the case that many health expenditures are unnecessary is the Dartmouth Atlas, which documents spending by Medicare, the program that covers health care for the elderly in the United States. The atlas is a map of the United States that shows extraordinary variation in health-care expenditures from place to place, a variation that is linked neither to medical needs nor to better outcomes. Indeed, there is a *negative* correlation between expenditures and the quality of outcomes.[20] The most plausible interpretation is that some doctors and hospitals are much more aggressive than others in ordering tests and treatments and that those additional expenditures produce little or no benefit and in some cases may actually harm patients. If this is true, health-care expenditures could be greatly reduced without harming health.

Given that health care is of high quality and helps maintain and improve health, it is an important instrument of wellbeing. But health care is expensive, so there is a potential trade-off between greater health care spending and other aspects of wellbeing. If Americans spent twice as much on health care, they would have to reduce expenditures on everything else by a quarter. Or if we could follow the Dartmouth recommendations in reducing expensive, low-value programs, and cut health-care expenditure by, say, half, we could have an increase of almost 10 percent in everything else. These sorts of trade-offs happen all the time in daily life, and we do not usually worry too much about whether people are, for example, spending too much on books or electronic gadgets so that they have too little left to spend on summer vacations. So why is health care different?

The problem is that people are not actually *choosing* how much to spend on health care in the way that they choose how much to spend on books or on vacations. Indeed people may not even be aware of what they are paying for health care, or what they are giving up to get it. In the United States, most health care for the elderly is paid for by the government through Medicare, and most (59 percent) of the non-elderly have coverage through their employers. Many of these think that their employers are *paying* for their health care at no cost to themselves. Yet most studies have shown that it is not employers that ultimately pay, for example through lower profits, but employees, through lower wages.[21] As a result, typical earnings, and the family incomes that depend on them, have grown more slowly than would have been the case if health-care costs had not grown so rapidly. But people do not see it that way, and do not think to blame rising healthcare costs for the slow growth in their incomes. As a result, they fail to see the cost of health care as the problem that it really is.

Similar problems arise when the government provides health care, as in Europe, or in Medicare, which pays for health care for the elderly in the United States. When people press for the government to provide additional health-care benefits—coverage for prescription drug benefits, for example—they tend not to think about what has to be given up in exchange. The doyen of American health economists, Victor Fuchs, gives the example of an elderly woman for whom Medicare will provide expensive surgery at no cost to herself, even surgery that might not be urgent or necessarily effective, but whose pension check is not enough to allow her to buy a plane ticket to attend her granddaughter's wedding or to visit a new grandchild.[22] These trade-offs have to be made through the political process by some sort of democratic debate, but that is a difficult, contentious, and often ill-informed process. It is also a process that, at least in some countries, is deeply influenced by the providers of health-care services, who have an interest in over-provision—an interest that becomes stronger and better financed the more that is spent.

Income and health are two of the most important components of wellbeing, and the two with which this book is mainly concerned. We cannot think about them one at a time, or allow doctors and patients to lobby for health improvements and economists to lobby for economic growth, each group ignoring the other. When health care is as expensive and as effective as it is today, trade-offs need to be made; in Fuchs's words, we must take a holistic view of wellbeing. Some process needs to be put into place to allow us to take that view collectively, almost inevitably involving an institution like Britain's NICE, as well as a greater and more widespread public understanding of the threats to other aspects of wellbeing that come from the unlimited growth of health-care costs.

What about the future? Can we expect life expectancy to continue its rise in the high-income countries? The negative view, often associated with the demographer and sociologist Jay Olshansky, starts from the observation that it is getting harder and harder to increase life expectancy. This is something we have already seen; saving children's lives has a dramatic effect on life expectancy, because they have so many years to live, but once nearly all of the children have been saved, saving the elderly makes less of a difference, at least to life expectancy. Figure 1 in Chapter 2 shows the distinct slowdown in the rate of increase in American life expectancy after 1950, and the argument is that we can expect a similar slowdown in the future, even if innovations continue, because the lives saved will be of ever-older people. Even if cancer were eliminated in the United States, life expectancy would increase by only four or five years. The pessimists also note that the rise in obesity in most rich countries may raise mortality rates in the future. Perhaps, but there has been little evidence of it so far. This may be because, with better treatments for cardiovascular disease—including drugs to control cholesterol and hypertension—the risks of obesity are lower now than when they were first studied.[23]

On the other side, the demographers Jim Oeppen and James Vaupel published in 2002 a remarkable diagram that calculated the world's highest life expectancy for women in each year from 1840, and showed that this measure—which can be thought of as the maximum possible life expectancy in each year—has risen at a constant rate for 160 years.[24] For every four years of calendar time, the world's highest life expectancy increased by a year. Oeppen and Vaupel see no reason why this long-established rate of progress should not continue. Their diagram also marks the many previous estimates of the maximum possible life expectancy, each of which was swept away by actual events; many previous sages have forecast that the gains to life span will slow or stop, and they have all been wrong. Further supporting the optimistic argument in favor of a continued rise in life expectancy is the fact that people do not want to die any earlier than they must; that as they get materially richer they have more income to spend trying to avoid that outcome and will likely be prepared to devote a larger and larger share of their incomes to staying alive; and there is no reason to suppose that they will not succeed in the future as they have in the past.

I find the optimistic argument the more compelling: ever since people rebelled against authority in the Enlightenment, and set about using the force of reason to make their lives better, they have found a way to do so, and there is little doubt that they will continue to win victories against the forces of death. That said, it is too optimistic to think that life expectancy in the future will grow at the same rate as it did in the past; falling rates of infant and child mortality make life expectancy grow rapidly, and that source of growth is largely gone, at least in the rich countries. During the 160 years when top life expectancy grew by a year every four years, a substantial contribution came from saving the lives of children, and that will not continue. Once again, there is good reason *not* to focus on life expectancy as the measure of success. Eliminating cancer and other diseases of the elderly would eliminate great suffering and improve millions of lives. That it would have a modest effect on life expectancy is largely beside the point.

Health in an Age of Globalization

I have looked at rich countries (in this chapter) and poor countries (in Chapter 3) as if they were separate worlds. Now it is time to look at them together, and to think about how the two groups affect one another. The past half-century has seen an unparalleled integration of the world—a process often referred to as globalization. This is certainly not the first instance of globalization in history, although the current episode is one of the most far reaching. Transport is faster and cheaper than ever before, and information moves faster still. Globalization has affected health in many ways: directly through the spread of disease, information, and treatment, and indirectly through economic forces, particularly increased trade and higher economic growth.

There have been many periods of globalization in history—sometimes through war, conquest, and imperialist expansion, sometimes through new trade routes, bringing new products and new riches. Disease usually came along for the ride, with consequences that reshaped the world. The historian Ian Morris has described how increased trade around the second century CE merged previously separate disease pools that, since the beginning of agriculture, had evolved in the West, South Asia, and East Asia, “as if they were on different planets.” Catastrophic plagues broke out in China and in the eastern outposts of the Roman Empire.[25] The Columbian exchange after 1492 is an even better-known example.[26] Many historical epidemics started from new trade routes or new conquests. The plague of Athens in 430 BCE was attributed to trade, and bubonic plague was brought to Europe in 1347 by rats aboard trading ships. The cholera epidemic of the nineteenth century is thought to have come from Asia thanks to the activities of the British in India, and its subsequent spread through Europe and North America was speeded by the new railways. An infected person could be in another city before he or she knew of the infection, and cholera spread along the railway lines; today, someone can move from one hemisphere to another in the time it used to take to go from one city to another.

Yet globalization also opens its routes to the enemies of disease. We have already seen how the germ theory of disease—a set of ideas and practices developed in the North—spread rapidly to the rest of the world after 1945. Knowledge about drugs to control high blood pressure spread rapidly across the world after 1970, producing the synchronized declines in mortality plotted in Figure 3. That cigarette smoking caused cancer did not have to be rediscovered country by country. While the origins of HIV/AIDS are in dispute, there is no dispute about its rapid spread from one continent to another. The scientific response—the discovery of the virus, the deduction of its means of transmission, and the development of chemotherapy that is transforming the disease from a fatal to a chronic condition—was extraordinarily rapid by historical standards, although hardly rapid enough for the millions who died as they waited. Today's understanding of the disease, although still incomplete, has underpinned the response—not just in the rich world—and in the worst affected African countries rates of new infection have fallen in the past few years, and life expectancy is beginning to rise again.

Successes against cardiovascular disease and cancer are spreading, not just from one rich country to another, but throughout the world. As mortality from infectious disease has fallen, noncommunicable diseases are becoming more important as the children who did not die become adults and live long enough to encounter them. Except in Africa, noncommunicable disease is now the leading cause of death everywhere in the world, and cheap and effective preventative drugs like antihypertensives should spread just as vaccines did in the past. Here, once again, the constraint is likely to be the capacity of some governments to organize and regulate a physician-based system of health care. Advances that are more expensive, such as some cancer treatments or joint replacement, are also spreading, but they are typically available only to the well heeled or well connected in a limited number of poor countries.

The contributions of rich-country to poor-country health have not always been benign. Health researchers, unlike economists, often think of globalization as a negative force. There is deep concern about cigarette smoking, and the activities of the tobacco companies whose products, no longer welcome in much of the rich world, are finding a safe haven in poorer countries whose governments, once again, may not have the capacity for or an interest in regulation. The patent system that makes drugs temporarily very expensive has been vociferously challenged, though it is not clear whether patents are the real problem. Again, local capacity for delivery is an issue and, in any case, nearly all of the drugs that the WHO lists as "essential medicines" are off patent; even so, the list might be longer if more drugs were cheaper. Small poor countries often find themselves at a disadvantage when negotiating bilateral trade deals with large rich countries. The latter are much better supplied with lawyers and lobbyists, including pharmaceutical lobbyists, whose interest is not to protect poor-country health. First-world medicine has certainly sharpened local health inequalities in poor countries. In cities like Delhi, Johannesburg, Mexico City, and São Paulo, first-world state-of-the-art medical facilities treat the wealthy and powerful, sometimes within sight of people whose health environment is not much better than that of seventeenth-century Europe.

What has happened to global health and global health inequalities since 1950? In Figure 1 of Chapter 3, we saw that regional inequalities in life expectancy have narrowed, so that the regions with the lowest life expectancy have drawn closer to the regions with the highest life expectancy. Now I look at countries, not regions, as units. Figure 4 shows how life expectancy is changing in the typical country, how the worst and best countries are doing, and whether inequality in life expectancy is getting larger or smaller. The graph looks like a series of organ pipes, though it as actually known as a "box-and-whisker" plot. The vertical axis shows life expectancy, and the pipes (or boxes) show where countries' life expectancies line up; the first message from the graph is that the pipes are rising from bottom left (1950–54) to top right (2005–09) as longevity increases around the world. Each shaded box contains half of all countries, and the line across the middle marks the middle country in terms of life expectancy. These horizontal lines are rising over time—life expectancy in the middle country is going up—albeit at a rate that is somewhat slower now than fifty years ago. Once again, the reason is that we have moved from the large increases in life expectancy that come from saving children to the smaller and more hard-won gains that come from saving the elderly. The top and bottom "whiskers," shown as lines coming out of the pipe with bars on the end, are designed to capture all countries except those that are really extreme in their longevity. The figure shows only two countries that are extreme in this way, both of which were in the midst of civil wars between 1990 and 1995: Rwanda and Sierra Leone. A total of 192 countries are plotted for each period, and some of the estimates are speculative, especially in the earlier years.

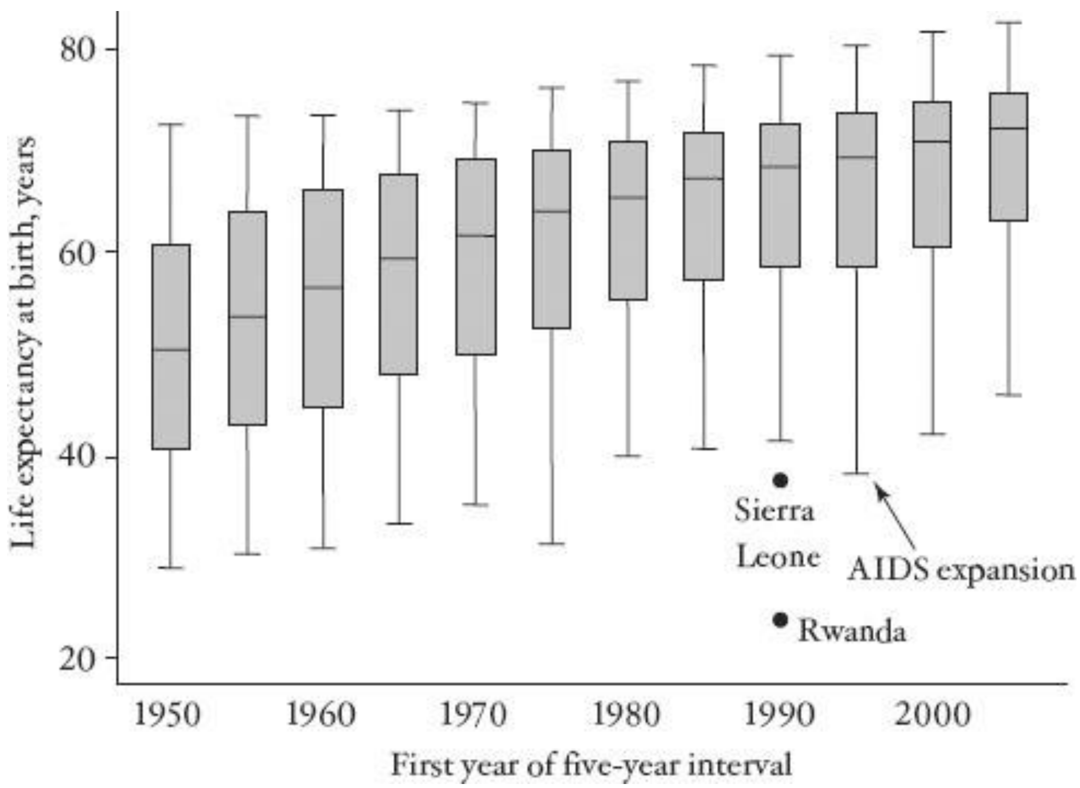

FIGURE 4 Life expectancy and its dispersion around the world.

The figure shows the pipes getting smaller over time, so that countries are moving toward the middle of the crowd. The dispersion of life expectancy across countries is narrowing, and this measure of the international distribution of health is becoming less unequal. The explosion of international health inequality that began 250 years ago is beginning to be reversed. The narrowing has not been entirely even, and we can see the widening in 1995–2000 that comes from AIDS deaths in Africa, after which the narrowing resumes. The bars in the middle of the boxes are getting progressively closer to the tops of the pipes, and to the top whisker, which tells us that the gap between life expectancy in the middle country and the top countries—like Japan—has also been narrowing over time. There is now only a 10.5-year gap between the middle—more precisely the median—(72.2 years) and the top (Japan at 82.7 years). Yet this narrowing is leaving a longer tail of countries behind it. Even ignoring the horrors in Rwanda and Sierra Leone in the early 1990s, the gap from middle to bottom has grown from 22 to 26 years.

Once again we must ask ourselves whether life expectancy is a good measure for thinking about health inequalities across countries. This chapter has shown that the gains have come from saving children in poor countries and from saving the middle-aged and elderly in rich countries. When we use gains in life expectancy to compare across rich and poor, we give greater weight to the poor countries, because saving a child has a much larger effect on life expectancy than saving a 60-year-old. And indeed, this is the main reason why inequality in life expectancy has declined. Yet it is not obvious that saving a child is indeed better than saving an older adult, a judgment that is built in to this measure of inequality. This point can be argued either way. Some would argue for saving the child—because, even if he or she does not yet have much stake in the world, so many future years are gained—and others for saving the adult—because he or she has a larger stake, albeit with fewer years left. But there is nothing that says that using life expectancy to look at inequalities solves this conundrum in just the right way, and weighing lives differently could make the reduction in inequality larger or smaller, or might even reverse it.

The reductions in global inequalities in life expectancy do not automatically mean that the world is a better place, because life expectancy does not capture all the aspects of health—or even of mortality—that we care about. Certainly, we live in a world in which child mortality is falling in poor countries, and middle-aged and elderly mortality is falling in rich countries. Whether those trends make the world a more equal place is a debatable issue that depends on how much we care about each kind of mortality decline.

The philosophical arguments do not stop there. The reductions in infant and child mortality have been followed by reductions in the number of children that people choose to have. In Africa in 1950, each woman could expect to give birth to 6.6 children; by 2000, that number had fallen to 5.1, and the UN estimates that it is 4.4 today. In Asia as well as in Latin America and the Caribbean, the decline has been even larger, from 6 children to just over 2. Fertility did not fall immediately after mortality rates fell, which is why there was an explosion in population. Yet in time, as parents came to believe that not so many children would die, they stopped having so many babies, though they may have had as many or more who survived to adulthood. One way of thinking about this change is that babies, who would have been born and quickly died, are now not being born at all. Who are the beneficiaries of this change? Again, it depends on how we choose to weigh lives, a question that has engaged philosophers for a long time. Yet it is clear that *mothers* benefit a lot; they do not have to be pregnant as often to have the same number of children who survive, and they—and their husbands—are spared the agony of having their children die. Lightening this burden on women not only removes a source of pain, it also frees them to lead fuller lives in other dimensions, becoming more educated, working outside of the home, and playing a fuller role in society.

Changing Bodies

There is much to cheer about in global health since 1950. But I want to conclude with a somewhat less cheery set of observations that focus not on the Great Escape from death—which has been impressive in most places, and perhaps even equalizing—but on the less impressive and less equal progress in the Great Escape from malnutrition. One good way to look at malnutrition is to see what has happened to human height.

Height, in and of itself, is not a measure of wellbeing. Other things equal, there is no reason to suppose that someone who is over six feet tall is any happier, richer, or healthier than someone who is six inches shorter. Nor is height a part of wellbeing in the way that income and health are parts of wellbeing. Yet when a *population* is short, it indicates that its members were nutritionally deprived in childhood or in adolescence, either because they did not get enough to eat, or because they lived in an unhealthy environment where disease, even if it did not kill them, left them permanently stunted. While height depends on genes for each individual, so that taller parents have taller children, it is now believed that this is not true for (sufficiently large) populations, and that variations in average height among populations are good indicators of variations in the degree of deprivation. In the past, we thought that genetic differences were the main source of differences in heights across populations. But as conditions improved, and one "short" country after another has grown taller, sometimes quite quickly, these views have been discarded.[27]

We are now beginning to understand that deprivation in childhood can have serious and long-lasting consequences. Shorter people earn less than taller people, not only in agricultural societies, where strength and physique are useful in the labor market, but also among professionals in rich countries such as Britain and the United States. One reason is that cognitive function develops along with the rest of the body, so that shorter people, *on average,* are not as smart as taller people—a statement that tends to call forth howls of outrage. Two of my Princeton colleagues who investigated this question[28] were denounced, bombarded with hate-filled emails, and subjected to demands by alumni that the university dismiss them. So let me try to explain carefully.

In an ideal environment where everyone gets enough to eat and where no one ever gets sick, some people will be short and some tall, according to their genetic makeup, but there will be *no systematic difference* in cognitive function according to height. In the actual sublunary world, some people will be deprived in childhood, and those people will be overrepresented among the short, which is why short people, on average, have poorer cognitive function. This may be simply a matter of insufficient calories, or of fighting off one too many childhood diseases, an important drain on calories. The deprivation can also be more specific; for example, children's brains need fat to develop properly, and there are millions of people in the world whose diets contain *too little* fat, in contrast to the more familiar millions whose diets contain too much.

Nutritional deprivation wears off as populations become richer and get enough to eat, and as childhood disease is banished through sanitary improvements, pest control, and vaccines. Even so, the effects of nutritional deprivation on heights may take many years to wear off, if only because tiny mothers cannot have large children. The rate of height increase in a population is subject to this biological limit, so that it may take many generations for populations to grow to their full potential, even after the nutritional and disease constraints have been removed; biology limits growth to avoid problems that would result from fast catch-up.[29] But over time we would expect to see the people of the world getting taller. It turns out that some have and some have not.

Europeans have become *much* taller. The economists Timothy Hatton and Bernice Bray have assembled data on men's heights from various sources for eleven European countries going back into the late 1850s or early 1860s.[30] Unfortunately there are very few historical data on the heights of *women*, because the information on men's heights typically comes from measurement during recruitment for armies. For those born in the middle of the nineteenth century, the height of the average European adult man was 166.7 centimeters (cm), or 5 feet 5.5 inches. For those born a little over a hundred years later, in the five years from 1976 to 1980, the average was 178.6 cm, or 5 feet 10.5 inches. In the country that grew the least—France—the rate of growth was 0.8 cm for each decade; in the fastest-growing—the Netherlands—the rate of growth was 1.35 cm for each decade. Men in most of the other countries grew taller by something close to 1 cm for every decade. Hatton has tracked these improvements back to their underlying causes and, in line with the arguments of this chapter, finds that the reduction in infant mortality—an indication of an improved disease environment—was the most important factor, with growth in income in second place.[31] As Europe made its escape from too little food, and the "cloacal infernos"[32] created by the Industrial Revolution, people's bodies began to grow toward the heights that had always been possible but were previously unachievable.

For most of the world today, there is only fragmentary historical information, but we do have good information on the heights of women from many of the Demographic and Health Surveys discussed in Chapter 2. (The most recent of these surveys measured men too.) Each survey gives us historical information because it measures people aged 15 to 49. Because people's heights do not change once they reach their adult height (or at least not until they start shrinking after age 50), each survey gives us the average adult heights of people born over a period of twenty or more years. So not only do these surveys give us the average heights of female adults in the country at the time of the survey, but by comparing older and younger women we can also see how fast heights are growing. In countries that are doing well, the older women will be a centimeter or two shorter than the younger women.

Figure 5 shows women's heights around the world. Each of the points in the figure is for a "birth cohort" of women for a country; it is the average height in centimeters of all the women born in a particular year, say 1960. That average is plotted against average national income in that country in the women's year of birth, and once again I have used a log scale for income. At the top right of the picture, for example, we see European women getting taller as their national incomes rose; women born earlier are at the bottom left of the European group, and women born later are at the top right. The United States is shown as a spur on the right; Americans have been growing taller but not as quickly as Europeans. In the middle and to the left of the diagram we see the women of poor and middle-income countries. The dark circles are for Africa and appear mostly to the left, because African countries were poor when these women were born, just as they are poor today. (The rich Africans on the right of the diagram live in Gabon, whose oil exports give it a high per capita income, although most of its inhabitants remain poor.) Nestled among the Africans are Haitians (white circles), most of whom are of African descent, and whose heights and incomes give them much in common with Africans in Africa. China (in gray) is also on the left, and Bangladesh, India, and Nepal are on the bottom at the left. Remember that we are looking at incomes in the years in which these now-adult women were born, typically 1980 or earlier, so that China and India show up as much poorer than they are today. Latin American and Caribbean women, who live in middle-income countries, appear in the lower middle part of the figure.

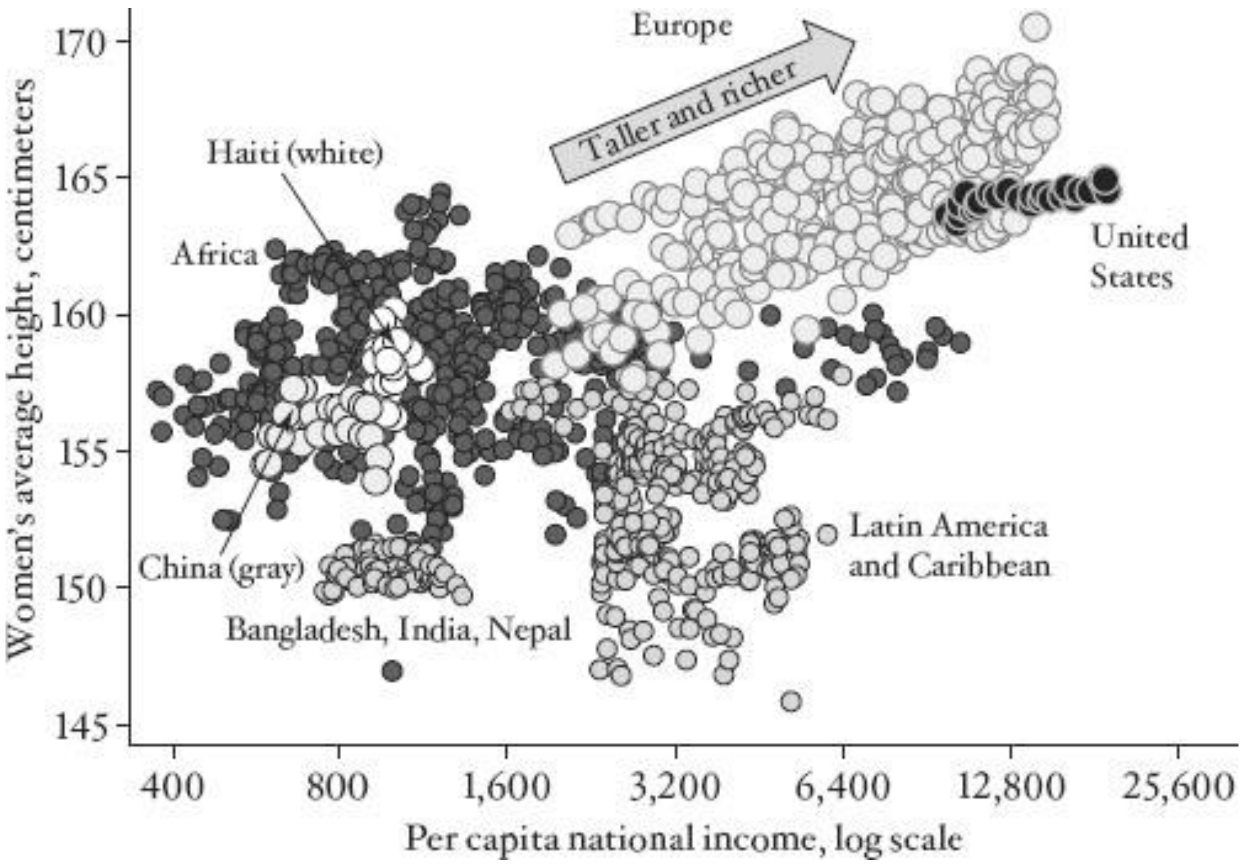

FIGURE 5 Women's heights around the world.

Perhaps the most startling feature of the diagram is the enormous inequality of average heights around the world. For women born in 1980, the average adult Dane was 171 cm, the average Guatemalan was 148 cm, the average Peruvian or Nepali 150 cm, and the average Indian, Bangladeshi, or Bolivian 151 cm. If the shortest populations in the world were to grow at the European rate of 1 cm every decade, it would take 230 years for the Guatemalan women to be as tall as Danish women are today. Today, a Danish woman visiting a Guatemalan village would tower 9 inches above her hosts, a modern Gulliver in a modern Lilliput.

Looking from bottom left to top right, people in better-off countries are taller than people in poorer countries, which is what we might expect if higher incomes come with better sanitation, lower rates of childhood disease, and more to eat. But things are not quite so simple. Imagine the figure redrawn with Europe and the United States blanked out. For the rest of the world, the relationship between height and income goes the wrong way, with the *taller* people living in the *poorer* countries. A lot of this has to do with Africa. There is a lot of variability across African populations—think of Dinka basketball players from southern Sudan or the bushmen of the Kalahari—but on average African women are tall, not relative to Europeans but relative to South Asians and many Latin Americans. This negative relationship between heights and incomes is not about to go away any time soon, because Indian children born today are *still* very short, in spite of the rapid growth of the Indian economy in recent decades.

Why Africans are so tall is not well understood. One reason is that, in much of the continent, food is neither so scarce, nor so heavily vegetarian, as in much of South Asia, particularly India. Obviously, this is not true in some places—the Kalahari Desert, for example—but in most African countries, people eat a varied diet that includes meat and animal fats. There is also great variability in Africa, depending on local food availability and local disease environments. At the same time, mortality rates among children are extremely high, and if the shorter children are weaker and more likely to die, the survivors will be relatively tall. For this to produce a tall population, mortality has to be very high, high enough to sweep away a large enough fraction of short children and to overcome the stunting effect of living through a dangerous disease environment in childhood. Sanitation may be another factor; in places where people defecate in the open, and where population density is high, child growth is stunted by chronic exposure to fecal germs. Africa, with its much lower population density, does better than India.[33]

The fact that residents of many African nations are taller than those of India, or of several countries in Latin America, should help us resist the superficially attractive idea that population average heights can be used as some overall measure of wellbeing or of the standard of living. Mortality and income are two of the most important influences on adult height, and they are also crucial for well-being. But there is no guarantee that the way that disease and poverty affect height is the same as the way that they affect wellbeing. And as the map of Africa attests, many local factors—such as variations in diet—affect height, and these local factors may or may not affect well-being. Recall too that it can take many generations for populations to grow taller, because the mothers have to grow before the children, the grandmothers before the mothers, and so on. It is not just today's nutrition and today's diseases that determine today's height; history matters too. All of this means that average height is not a sensible measure of wellbeing.

The fact that South Asians are so short is perhaps the most informative part of the whole picture. Because we do not have historical data on European women, we do not know how far we have to go back to get to modern Indian heights. However, the latest Indian data include men, and it turns out that the average height of Indian men born in 1960 was 164 cm: 2–3 cm shorter than the European average in 1860, similar to European heights in the eighteenth century, and only 5 cm taller than the lowest figures in the literature, 159 cm from contemporary bushmen and from Norway in 1761.[34] In Sikkim and Meghalaya, states in northeastern India, the average heights of men born in 1960 are actually less than 159 cm.

It is possible that the deprivation in childhood of Indians born around midcentury was as severe as that of any large group in history, all the way back to the Neolithic revolution and the hunter-gatherers that preceded them. Life expectancy in India in 1931 was 27, also reflecting extreme deprivation. Indians, even in the twentieth century, lived in a Malthusian nightmare. As in Malthus, death and deprivation kept the population in check, but even for the survivors, the conditions of life were terrible. Not only was there insufficient food to maintain good health, but its makeup also lacked important nutrients. Most people ate a monotonous diet of a single cereal, supplemented by a few vegetables, lacking iron and adequate fat. In order to survive at all, even with a life expectancy in the 20s, the whole population had to be short, just as were the populations of England in the seventeenth and eighteenth centuries. The Malthusian imperative trades off more people for shorter people.

India is today escaping from this nightmare, but it still has a long way to go. Indian children are still among the skinniest and shortest on the planet, but they are taller and plumper than were their parents or grandparents, and the signs of gross hunger, such as marasmus, are now rarely seen in nutritional surveys. Indians too are now growing taller, decade by decade, though not as quickly as happened in Europe, or indeed as is now happening in China, where people are growing at about (the now familiar figure of) a centimeter every decade. Yet the Indian escape is only half as fast—about *half* a centimeter a decade—and that figure is for *men;* Indian women are growing too, but at a much slower rate, so that it takes them sixty years to grow a centimeter.[35]

We do not know why Indian women are doing so much worse than Indian men; the reason is surely linked to general patterns of favoring sons in north India, though exactly how this works is not known. In south India, in Kerala and Tamil Nadu, which have no tradition of bias against girls, both men and women are growing taller at the standard centimeter-a-decade rate, but in the north, women are growing more slowly than men, who are themselves growing less fast than their counterparts in the south. An irony of this sort of discrimination against women is that it rebounds against the men, because men, just like women, have unduly small and undernourished women as mothers, compromising their own prospects for physical and cognitive development.

In Africa, although people are taller on average, women in some places are actually getting *shorter.*[36] Although, as we have seen, it is not always true that better-off people are taller, there is a strong correlation around the world between getting richer and getting taller. This is most obviously true in Europe, and the growth has been sufficiently prolonged for it to be visible in Figure 5, but it is also true in modern China, India, and elsewhere. So the most likely reason that African women are shorter than their mothers is falling real incomes in Africa in the 1980s and early 1990s.

The world's peoples are not just living longer or getting richer; their bodies are getting taller and stronger, with many good consequences, including even possibly an increase in cognitive ability. But as with mortality and money, the distribution of benefits has been unequal. At current rates, it will take centuries for the Bolivians, Guatemalans, Peruvians, or South Asians to become as tall even as Europeans are today. So while many have made their escape, millions more are left behind, resulting in a world of difference in which inequality is apparent even in people’s bodies.

PART II **MONEY**

FIVE

Material Wellbeing in the United States

STARTING IN THE MIDDLE of the eighteenth century in Britain, longevity slowly began to improve in countries around the world. As people made their escape from disease and early death, living standards began to improve too, and, to a large extent, health and the level of living moved in parallel. The ideas of the Scientific Revolution and the Enlightenment eventually brought a revolution in material well-being, just as they brought a revolution in the length of life. These parallel revolutions, driven by the same ultimate causes, brought better and longer lives for many, but also created a world of difference through what the economist Lant Pritchett has memorably called "Divergence, big time."[1] Economic growth brought better living standards as well as a reduction in poverty. It is hard to measure any of this precisely—a situation about which I will have more to say—but one careful study estimates that the average income of all the inhabitants of the world increased between seven and eight times from 1820 to 1992.[2] At the same time, the fraction of the world's population in extreme poverty fell from 84 to 24 percent. This historically unprecedented increase in living standards came with huge increases in income inequality, both between countries and between individuals within countries. The *nature* of inequality also changed. In the eighteenth century, most inequality was *within* countries, between the rich land-owning aristocrats on the one hand and the common people on the other. By the year 2000, by contrast, the biggest gaps were *between* countries, the end result of the "big-time" divergence. In contrast to the shrinking gaps in longevity that we have already seen, the income gaps between countries show no sign of diminishing today.

I start with the material wellbeing of the United States, focusing on the past hundred years. I have chosen the United States because the story is a dramatic one and because it illustrates the central themes of this book. When wellbeing improves, not everyone benefits equally, so that the improvement often (although not always) widens gaps between people. Change, positive or negative, is often unjust. What happens to inequality matters not only for how we should judge the improvement—who is reaping the benefit and who is being left behind—but also because inequality has its own effects. Inequality can sometimes spread growth, if it shows a way for others to benefit from the new opportunities. But it can also undercut material improvement and even threaten to extinguish it altogether. Inequality can inspire or incentivize those who are left behind to catch up, generating improvements for them and for others. Or inequality can become so severe, and the gains so concentrated in the hands of a few, that economic growth is choked off and the workings of the economy are compromised.

I have also chosen to start with the United States because the data are good and are easy to understand. Everyone knows what a dollar is, we do not need to convert currencies, and we can rely on data provided by a first-class statistical system. None of these luxuries will be available when we come to look at the world as a whole. Similarly, when we look far back in time, the data are weaker and the basis for comparison more tenuous. Comparing the twenty-first and nineteenth centuries is in many ways as difficult as comparing two different countries. The people are different, what they spend their money on is different, and the standards of value are different; "the past is a foreign country." The ease of working with data from the United States gives me a familiar environment in which I can also develop some concepts and try to clarify what economists and statisticians mean when they talk about and try to measure income, poverty, and inequality.

Economic Growth in the United States

The familiar concept of gross domestic product (GDP) is a good place to start (though it would be a very poor place to stop). The top line of Figure 1 shows what has happened to GDP per capita in the United States since 1929, when the modern statistics begin. GDP is a measure of how much a nation produces and is the basis for national income. It was just over $8,000 per head per year in 1929, fell to $5,695 in 1933 in the depths of the Great Depression, and increased—with a few hiccups—to $43,238 in 2012, a more than fivefold increase since 1929. These numbers are corrected for increases in prices over time, so that they are measures of *real* income per head, measured in dollars at the prices of 2005. The figure for 1929 tells us that average national income in 1929, which was $805 in the much lower prices of the time, would have been worth $8,000 in 2009.[3]

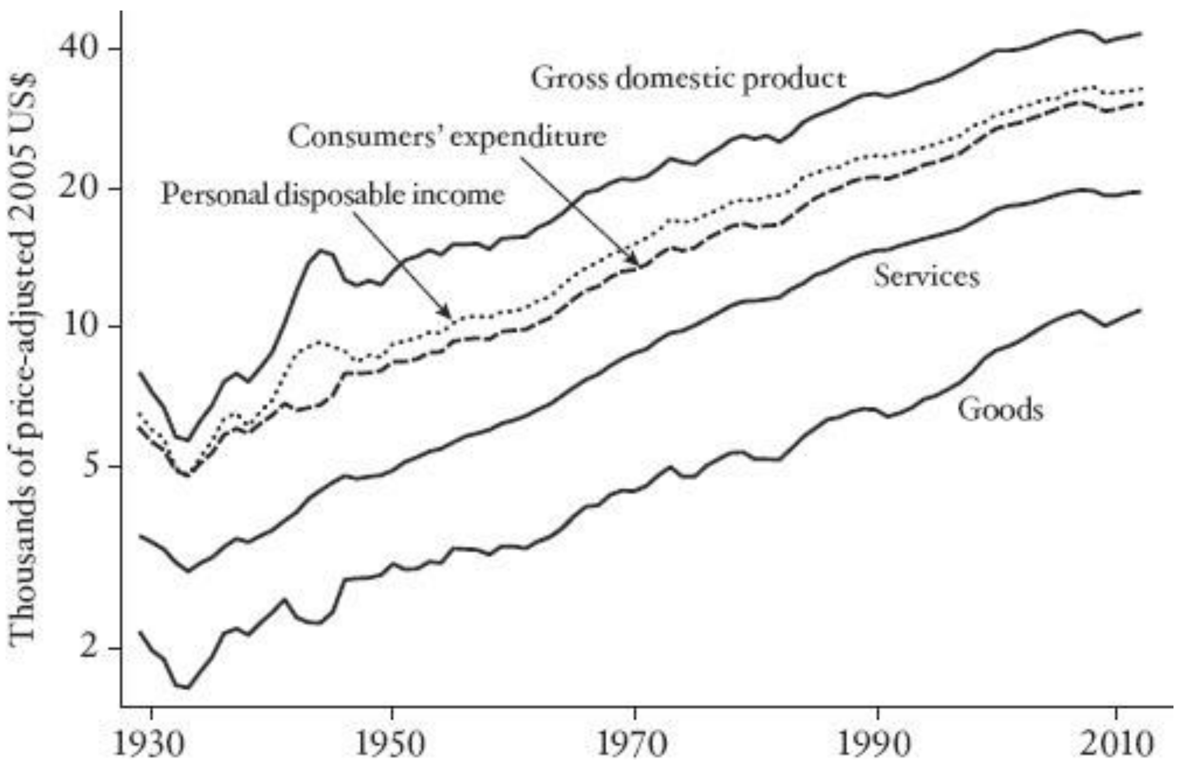

FIGURE 1 Gross domestic product and its components, 1929–2012.

The hiccups in GDP are times when progress stopped or was reversed; such hiccups have become less frequent and less severe over time—itself a measure of progress. The Great Recession that followed the financial crash of 2008 barely shows up in this history, in spite of the misery that it has caused, especially for the millions who became unemployed and remain so as I write. After 1950, the graph is close to being a straight line, which means a constant rate of growth at 1.9 percent per year, or a little more than 2 percent a year if we stop in 2008. Although the data get shakier as we move back in time, the growth rate of per capita national income has not changed very much over the past century and a half. At 2 percent growth, income doubles every thirty-five years, so if each couple had two children at age 35, each generation would have double the living standards of its parents. For those of us who are alive today, this seems like the natural order of things, yet it would have astonished our ancestors, who for thousands of years saw no progress at all, or who saw progress lost in subsequent setbacks. For all we know, it might also surprise our children and grandchildren.

As we shall see, GDP is a poor indicator of wellbeing, but it is limited even as a measure of income. It includes income generated in the United States that belongs to foreigners; it includes incomes in the form of undistributed corporate profits (which ultimately belong to shareholders) as well as surpluses run by federal, state, and local government. The part of national income that is available to families, after taxes have been paid and any transfers received, is *personal disposable income,* which is the second line from the top. It is a good deal smaller than GDP, but the historical picture of growth and fluctuation is very similar. Much the same is true if we look, not at what people get, but at what they spend. This is *consumers' expenditure,* the third line. The difference between personal disposable income and consumers' expenditure is the amount that people save, and the figure shows that the fraction of their income that Americans save has been falling, especially over the past thirty years. We don't know exactly why this has happened, and there are several possible explanations: it is easier to borrow than it used to be; it is no longer as necessary as it once was to save up to make the deposit on a house, a car, or a dishwasher; Social Security has perhaps reduced the need to save for retirement; and the average American benefited from increases in the stock market and in house prices—at least until the Great Recession.

Capital gains can be cashed in and spent, or they can be used to build wealth even when people are not saving. In the economists' lexicon, saving is defined as the *difference* between income and consumption, both of which are flows of money per unit of time. *Wealth* is not a flow but a *stock*, the total in a ledger at a moment of time. It is increased by capital gains and decreased by capital losses—many Americans lost around half of their wealth after the financial crash in 2008; it also increases when people save and decreases when they "dissave," spending more than they earn, for example in retirement or during a temporary period of unemployment.

The figure also shows what people spent their money on, split into two broad categories, goods (more than a third of the total in 2012) and services. The largest two items in services are housing and utilities, currently about $2 trillion a year or 18 percent of total consumers' expenditure, and health care, running at about $1.8 trillion, or 16 percent of the total. About a third of expenditure on goods is on durable goods—motor vehicles, furniture, electronics, and the like—and two-thirds is on nondurable goods—things like food and clothing. Americans today spend only 7.5 percent of their budget on food, or 13 percent once we add in expenditures on food consumed away from home. These expenditures are the stuff of material wellbeing, and their growth in Figure 1—and for the previous century—tells the tale of the increasing material prosperity that has accompanied increasing life spans. Life is not only longer, it is better.

Material wellbeing, and measures of it—GDP, personal income, and consumption—have recently received a bad press. Spending more, we are often told, does not bring us better lives, and religious authorities regularly warn against materialism. Even among those of us who endorse economic growth, there are many critics of GDP as it is currently defined and measured. GDP excludes important activities, such as services by homemakers; it takes no account of leisure; and it often does a poor job of measuring those things that *are* included. It also includes things that arguably should be excluded, like the cost of cleaning up pollution or building prisons or commuting. These "defensive" expenditures are not good in and of themselves but are regrettably